K-FIELD FOURIER GEOMETRY HIDDEN ARCHITECTURE

A foundational textbook reference for ternary affine geometry, projective incidence, complex Fourier modes, exterior duality, binary-ternary registry, exact phase sections, and dimensional collapse

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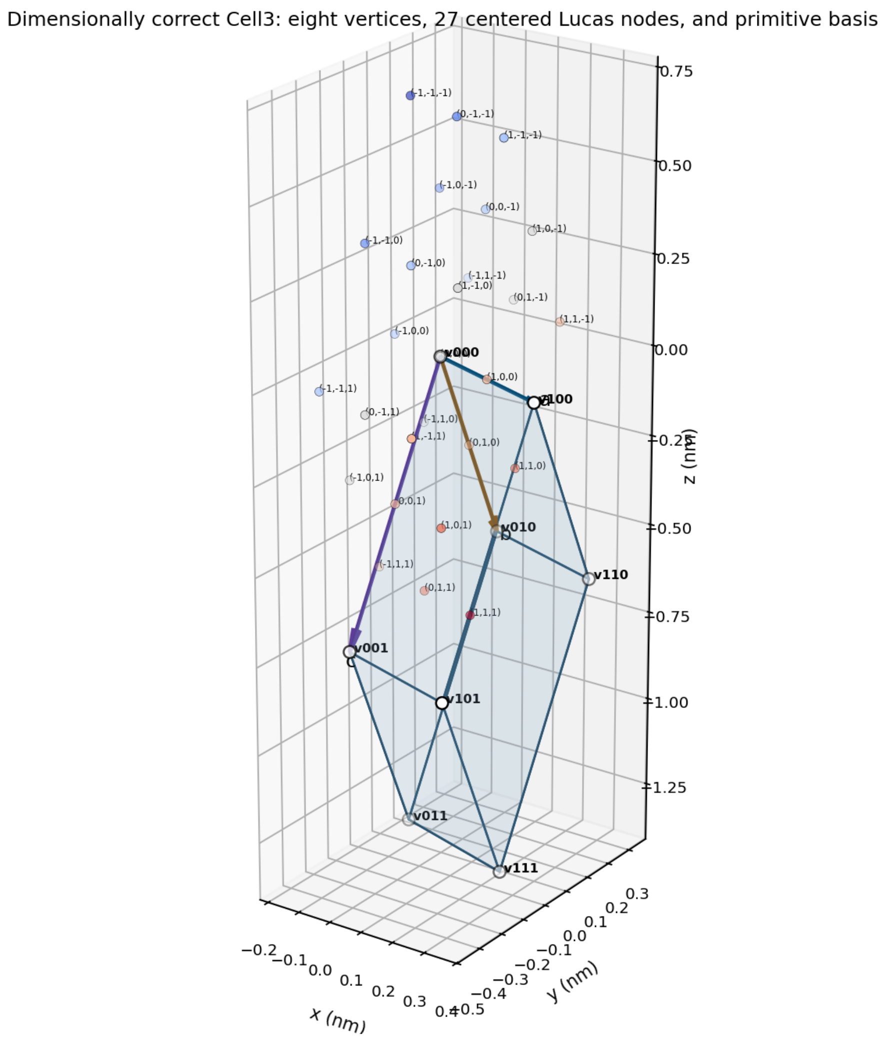

Title figure. Dimensionally correct Cell3 generated from the authoritative basis in nanometers. The eight boundary vertices, all 27 centered Lucas nodes, the six exterior faces, and the three primitive basis vectors are computed directly from the source record. Every Lucas node is labeled by its ternary address.

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Abstract

This reference develops the complete hidden architecture of Cell3 as a ternary affine Fourier geometry. The construction begins with the 27 addresses of the affine space AG(3,3), identifies the 13 antipodal directions of the projective plane PG(2,3), and then shows that those same 13 directions are the conjugate Fourier-mode pairs of every real field defined on the cell. The direct basis matrix physically embeds the finite geometry, while its inverse transpose supplies reciprocal plane normals and exact gradient-response magnitudes. Exterior algebra connects the reciprocal metric to the three face bivectors, so the Mass Bridge ratios become axial samples of one common grade-2 spectrum rather than independent appended constants.

The seven-sheet representation is reduced to an A1 plus A2 decomposition with an index-three gluing congruence. The sheet coordinate lambda controls both ternary phase and binary polarity. Their combined residue is a Z6 registry cycle, and the seventh visible sheet closes the six-step cycle. The phase tensor is shown to possess two exact circular-section planes whose common radius, normals, and projective angle are determined solely by its three eigenvalues. The solar-system ledger is likewise reduced: seven logged observables occupy only three dominant eigen-dimensions plus a small eccentricity coordinate because four variables are algebraically derived from primitive mass, scale, period, and eccentricity data.

The resulting architecture replaces an apparently heterogeneous collection of nodes, planes, shell radii, sheet states, phase axes, scale laws, and orbital observables with a compact hierarchy of finite geometry, Fourier representation, exterior metric response, binary-ternary recursion, and closure-null identities. The report provides exact equations, dimensionally correct figures, complete ledgers, worked derivations, and direct experimental pathways.

Central result. Cell3 is an obliquely embedded AG(3,3). Antipodal quotienting produces PG(2,3), whose 13 rays are simultaneously projective directions, reciprocal plane normals, conjugate Fourier-mode pairs, and two-dimensional real phase subspaces. The normalized reciprocal metric has five independent parameters; the 13 shell radii therefore obey seven exact closure-null relations.

Principal conclusions

- **Finite substrate:** 27 affine ternary points reduce to 13 antipodal projective rays plus the center.
- **Fourier identity:** every real 27-node field decomposes into one constant mode and 13 real two-dimensional conjugate-mode sectors.
- **Projective incidence:** h dot k = 0 mod 3 is exactly the condition that mode h has zero phase derivative along direction k.
- **Exterior rendering:** reciprocal radii are normalized magnitudes of coherent sums and differences of the three Cell3 face bivectors.
- **Registry closure:** lambda mod 3 supplies ternary phase, lambda mod 2 supplies polarity, and together they form a six-state cycle closed by seven displayed sheets.
- **Phase geometry:** the phase ellipsoid contains two exact circular sections and a six-node antipodal intersection skeleton.
- **Dimensional collapse:** seven orbital variables reduce to four primitive coordinates and an effectively three-dimensional observed manifold.

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1. Foundations, coordinate conventions, and authoritative data

Cell3 is described by a real 3 x 3 basis matrix B whose columns are the primitive vectors a, b, and c. All physical coordinates in the source record are expressed in nanometers. Integer addresses specify positions in the native lattice; multiplication by B renders those addresses in physical Cartesian space. The direct metric is G = B^T B. It records all squared lengths and mutual dot products of the primitive vectors. The reciprocal metric is G^(-1), and the physical reciprocal basis is B^(-T).

The distinction between address space and physical space is essential. Address space contains exact ternary and integer relations. Physical space supplies lengths, areas, angles, volumes, gradients, and anisotropic response. A finite-geometric statement remains true under every nonsingular change of B, whereas a metric statement changes with B. This separation makes it possible to identify which findings are topological, combinatorial, algebraic, or physically dimensioned.

The centered Lucas window uses addresses u = (i,j,k) with each coordinate in {-1,0,1}. Its physical coordinate is x = B u / 2. The factor one-half centers a doubled base-cell construction and reproduces every stored Lucas coordinate to machine precision. Boundary vertices use binary addresses in {0,1}^3 and physical coordinates B u without the factor one-half. The result is one dimensionally correct base parallelepiped containing the centered 27-node registry.

```
B = [a b c]
G = B^T B
G* = G^(-1)
x_vertex = B u, u in {0,1}^3
x_Lucas = (1/2) B u, u in {-1,0,1}^3
```

1.1 Primitive basis and dimensional invariants

Vector	x (nm)	y (nm)	z (nm)	Length (nm)
a	0.327501242069651	-0.054897007886313	-0.037487502896305	0.334179679073151
b	0.035330775844051	0.207319850441498	-0.547401085391957	0.586410890414836
c	-0.010129317626665	-0.389787702363804	-0.689734408362922	0.792319765037693

Metric record	Column 1	Column 2	Column 3
G row 1	0.111676057905435	0.020710333281606	0.043937235097789
G row 2	0.020710333281606	0.343877732397120	0.296392758961523
G row 3	0.043937235097789	0.296392758961523	0.627770610069385
G^(-1) row 1	9.208025528683773	0.001534802428118	-0.645188035676895
G^(-1) row 2	0.001534802428118	4.903398312306197	-2.315175584416388
G^(-1) row 3	-0.645188035676895	-2.315175584416388	2.731171274746109

The signed determinant is -0.117914891916699 nm^3, so the selected basis order is left-handed. The physical cell volume is |det B| = 0.117914891916699 nm^3. The three face areas are 0.357809539128562, 0.261106235799531, and 0.194869165469900 nm^2.

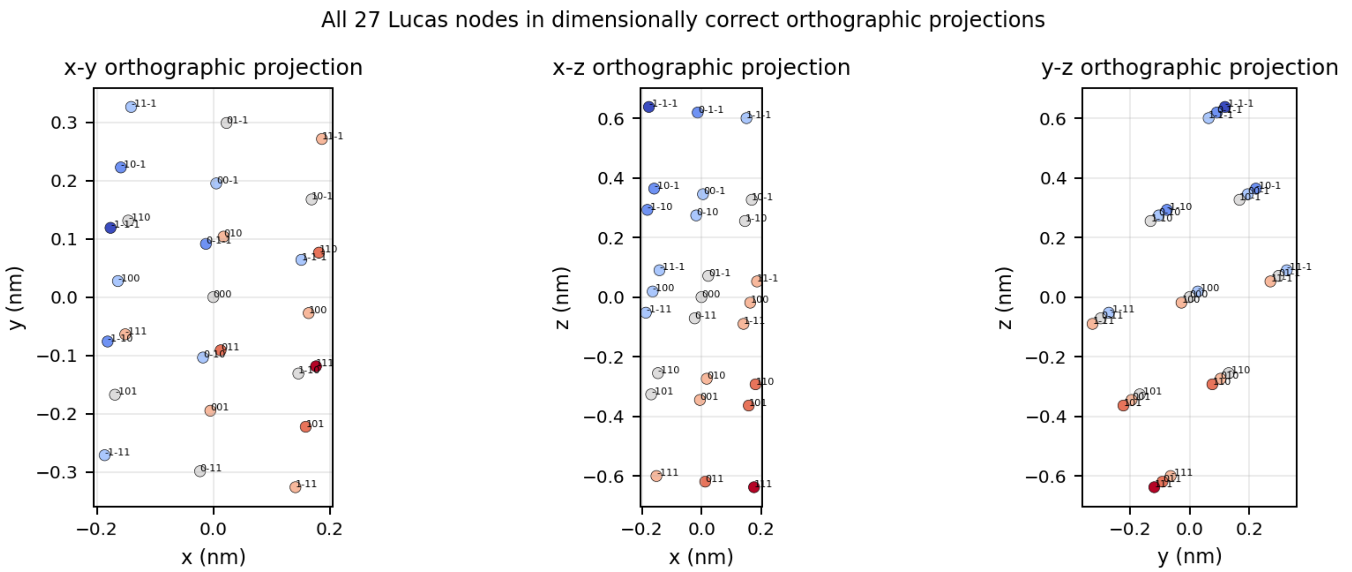

Figure 1. Orthographic projections of all 27 centered Lucas nodes. Coordinates and separations are generated from the source basis; node labels are compact ternary addresses. These views remove perspective ambiguity and make the finite 3 x 3 x 3 registry visible in each projected plane.

1.2 Complete 27-node Lucas ledger

The following table is the complete centered registry. No node is omitted. lambda, xi1, xi2, and polarity are computed from the integer address; the Cartesian coordinates and radii are physical quantities in nanometers.

Label	Address	lambda	xi1	xi2	chi	x nm	y nm	z nm	|x| nm
(-1,-1,-1)	(-1,-1,-1)	-3	0	0	-1	-0.176351350	0.118682430	0.637311498	0.671826811
(-1,-1,0)	(-1,-1,0)	-2	0	-1	1	-0.181416009	-0.076211421	0.292444294	0.352482076
(-1,0,-1)	(-1,0,-1)	-2	-1	1	1	-0.158685962	0.222342355	0.363610956	0.454785977
(0,-1,-1)	(0,-1,-1)	-2	1	0	1	-0.012600729	0.091233926	0.618567747	0.625386652
(-1,-1,1)	(-1,-1,1)	-1	0	-2	-1	-0.186480668	-0.271105272	-0.052422910	0.333198544
(-1,0,0)	(-1,0,0)	-1	-1	0	-1	-0.163750621	0.027448504	0.018743751	0.167089840
(-1,1,-1)	(-1,1,-1)	-1	-2	2	-1	-0.141020574	0.326002280	0.089910413	0.366398924
(0,-1,0)	(0,-1,0)	-1	1	-1	-1	-0.017665388	-0.103659925	0.273700543	0.293205445
(0,0,-1)	(0,0,-1)	-1	0	1	-1	0.005064659	0.194893851	0.344867204	0.396159883
(1,-1,-1)	(1,-1,-1)	-1	2	0	-1	0.151149892	0.063785422	0.599823995	0.621855044
(-1,0,1)	(-1,0,1)	0	-1	-1	1	-0.168815280	-0.167445347	-0.326123453	0.403600111
(-1,1,0)	(-1,1,0)	0	-2	1	1	-0.146085233	0.131108429	-0.254956791	0.321765879
(0,-1,1)	(0,-1,1)	0	1	-2	1	-0.022730047	-0.298553776	-0.071166661	0.307759169
(0,0,0)	(0,0,0)	0	0	0	1	0.000000000	0.000000000	0.000000000	0.000000000
(0,1,-1)	(0,1,-1)	0	-1	2	1	0.022730047	0.298553776	0.071166661	0.307759169
(1,-1,0)	(1,-1,0)	0	2	-1	1	0.146085233	-0.131108429	0.254956791	0.321765879
(1,0,-1)	(1,0,-1)	0	1	1	1	0.168815280	0.167445347	0.326123453	0.403600111
(-1,1,1)	(-1,1,1)	1	-2	0	-1	-0.151149892	-0.063785422	-0.599823995	0.621855044
(0,0,1)	(0,0,1)	1	0	-1	-1	-0.005064659	-0.194893851	-0.344867204	0.396159883
(0,1,0)	(0,1,0)	1	-1	1	-1	0.017665388	0.103659925	-0.273700543	0.293205445
(1,-1,1)	(1,-1,1)	1	2	-2	-1	0.141020574	-0.326002280	-0.089910413	0.366398924
(1,0,0)	(1,0,0)	1	1	0	-1	0.163750621	-0.027448504	-0.018743751	0.167089840
(1,1,-1)	(1,1,-1)	1	0	2	-1	0.186480668	0.271105272	0.052422910	0.333198544
(0,1,1)	(0,1,1)	2	-1	0	1	0.012600729	-0.091233926	-0.618567747	0.625386652
(1,0,1)	(1,0,1)	2	1	-1	1	0.158685962	-0.222342355	-0.363610956	0.454785977

Label	Address	lambda	xi1	xi2	chi	x nm	y nm	z nm	|x| nm
(1,1,0)	(1,1,0)	2	0	1	1	0.181416009	0.076211421	-0.292444294	0.352482076
(1,1,1)	(1,1,1)	3	0	0	-1	0.176351350	-0.118682430	-0.637311498	0.671826811

2. Cell3 as the affine ternary space AG(3,3)

The set {-1,0,1} becomes the three-element finite field F3 after reducing every coordinate modulo three. Addition and multiplication are performed modulo three; -1 is the same residue as 2. The 27 Lucas addresses therefore form the three-dimensional affine space AG(3,3). This is not a visual analogy. The point count, line count, plane count, translation law, and character group all agree exactly with the finite affine space.

AG(3,3) contains 27 points, 117 affine lines, and 39 affine planes. Every affine line has three points. Every affine plane has nine points. For each nonzero normal h, the equations h dot u = 0, 1, and 2 mod 3 define three parallel nine-point planes. Antipodal normals h and -h define the same parallel family, so the 26 nonzero normals reduce to 13 plane families.

The center u = (0,0,0) is fixed by sign reversal. Every other address forms an antipodal pair {u,-u}. The count is therefore 27 = 1 + 2 x 13. Quotienting the nonzero addresses by antipodal equivalence gives PG(2,3), the projective plane over F3. Its 13 projective points correspond one-to-one with the 13 reciprocal shell states in the source record.

```
AG(3,3) = F3^3, |AG(3,3)| = 3^3 = 27
PG(2,3) = (F3^3 - {0}) / {+/-1}
|PG(2,3)| = (3^3 - 1)/(3 - 1) = 13
27 = 1 + 2(13)
```

Hidden axiom 1. The 13-state shell is forced by the affine ternary cell. It is not an arbitrary catalog of preferred directions.

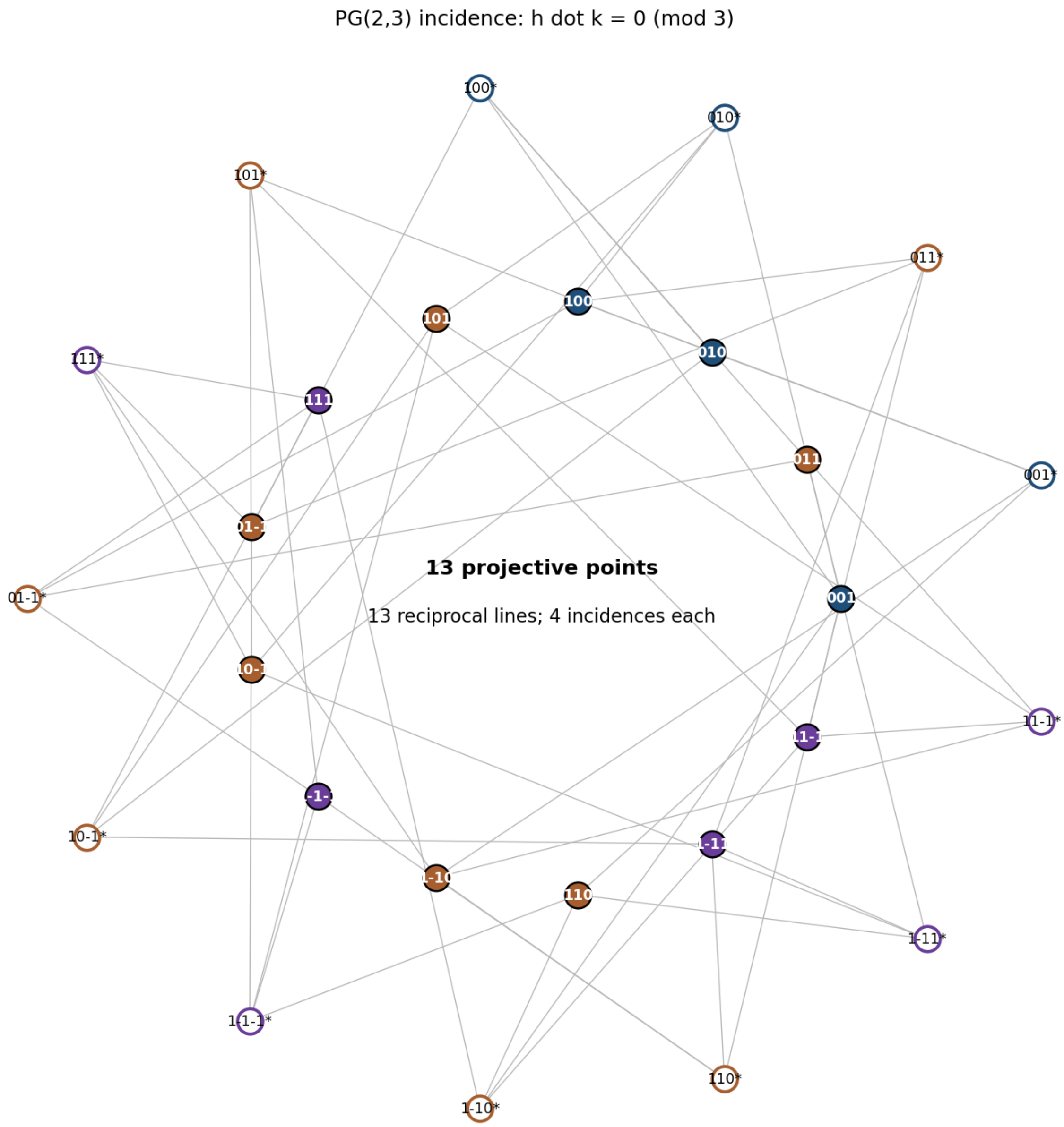

Figure 2. Exact point-line incidence graph of PG(2,3). Filled inner nodes are direct projective directions. Open outer nodes are reciprocal lines. An edge is drawn when h dot k = 0 mod 3. Every point lies on four lines and every line contains four points.

2.1 Projective classes and the finite quadratic form

Define q(h) = h1^2 + h2^2 + h3^2 mod 3. Every nonzero coordinate squares to one in F3, so q(h) is simply the number of nonzero coordinates reduced modulo three. This partitions the 13 projective states into three axial points, six face-diagonal points, and four body-diagonal points.

The body-diagonal quartet has q(h)=0. These four states are self-incident because h dot h = 0 mod 3. In projective language they form the four absolute points of the orthogonal polarity, which is a nondegenerate conic in PG(2,3). Lines associated with axial normals are external to the conic, face-diagonal normals are secants, and body-diagonal normals are tangents.

This conic is the smallest distinguished directional subset of the architecture. It separates self-propagating or self-invariant directions from the remaining shell. Any search for dynamically privileged axes should therefore test the body-diagonal quartet before treating all 13 states as equivalent candidates.

Code	h	Class	q mod 3	Self-incident	r(h)
001	(0, 0, 1)	axial	1	no	1.000000000000
011	(0, 1, 1)	face diagonal	2	no	1.048796632255
010	(0, 1, 0)	axial	1	no	1.339905341976
100	(1, 0, 0)	axial	1	no	1.836152673337
101	(1, 0, 1)	face diagonal	2	no	1.974587074999
111	(1, 1, 1)	body diagonal	0	yes	2.000023101807
01-1	(0, 1, -1)	face diagonal	2	no	2.119131490749
10-1	(1, 0, -1)	face diagonal	2	no	2.200890538590
1-1-1	(1, -1, -1)	body diagonal	0	yes	2.223234046047
1-10	(1, -1, 0)	face diagonal	2	no	2.272813025763
110	(1, 1, 0)	face diagonal	2	no	2.273307476001
1-11	(1, -1, 1)	body diagonal	0	yes	2.718195812787
11-1	(1, 1, -1)	body diagonal	0	yes	2.887171860683

3. The complete complex Fourier architecture

Every scalar field f on the 27-node cell is a function from F3^3 to the complex numbers. The additive group F3^3 has exactly 27 irreducible characters. Let omega = exp(2 pi i / 3). For each frequency address h, define psi_h(u) = omega^(h dot u). The 27 characters are mutually orthogonal and form a complete basis for every complex field on Cell3.

The zero address gives the constant character. Every nonzero h is paired with -h, and psi_-h is the complex conjugate of psi_h. A real-valued field therefore has one one-dimensional constant sector and 13 two-dimensional real sectors. Each sector is spanned by the cosine-like and sine-like parts of one conjugate character pair. The dimension count is exact: 1 + 13 x 2 = 27.

The projective shell is consequently also the complete nonconstant Fourier-mode registry. A projective state does not merely identify a ray in a geometric drawing. It identifies one independent real two-dimensional phase sector of every field that can exist on the 27-node cell.

```
omega = exp(2*pi*i/3)
psi_h(u) = omega^(h dot u)
psi_-h = conjugate(psi_h)
R^27 = R(constant) + direct_sum_[h in PG(2,3)] R^2
1 + 13(2) = 27
```

Hidden axiom 2. Projective direction and Fourier mode are the same finite object viewed in spatial and spectral representations.

3.1 Exact discrete derivative and physical gradient

A centered difference along address coordinate e_j compares the field one ternary step forward and one step backward. Acting on a character, this difference multiplies the character by i sin(2 pi h_j/3). Since h_j can only be -1, 0, or 1, the sine factor is exactly (sqrt(3)/2) h_j. The derivative is therefore linear in the finite frequency address.

The address derivative becomes a physical Cartesian gradient through B^(-T). The resulting gradient magnitude is governed by h^T G^(-1) h. After normalizing to the 001 state, this quantity reproduces every stored projective shell radius. The shell spectrum is therefore an exact discrete-gradient spectrum.

The direct metric determines lengths of position increments. The reciprocal metric determines magnitudes of phase gradients. This is the finite-cell analogue of the familiar inverse relation between wavelength and wave number, but here the allowed wave numbers are exactly the 13 projective states.

```
delta_j psi_h = [psi_h(u+e_j)-psi_h(u-e_j)]/2
= i sin(2*pi*h_j/3) psi_h
= i (sqrt(3)/2) h_j psi_h

grad_B psi_h = i (sqrt(3)/2) B^(-T) h psi_h
```

```
|grad_B psi_h|^2 = (3/4) h^T G^(-1) h |psi_h|^2
r(h) = sqrt[h^T G^(-1) h / (e3^T G^(-1) e3)]
```

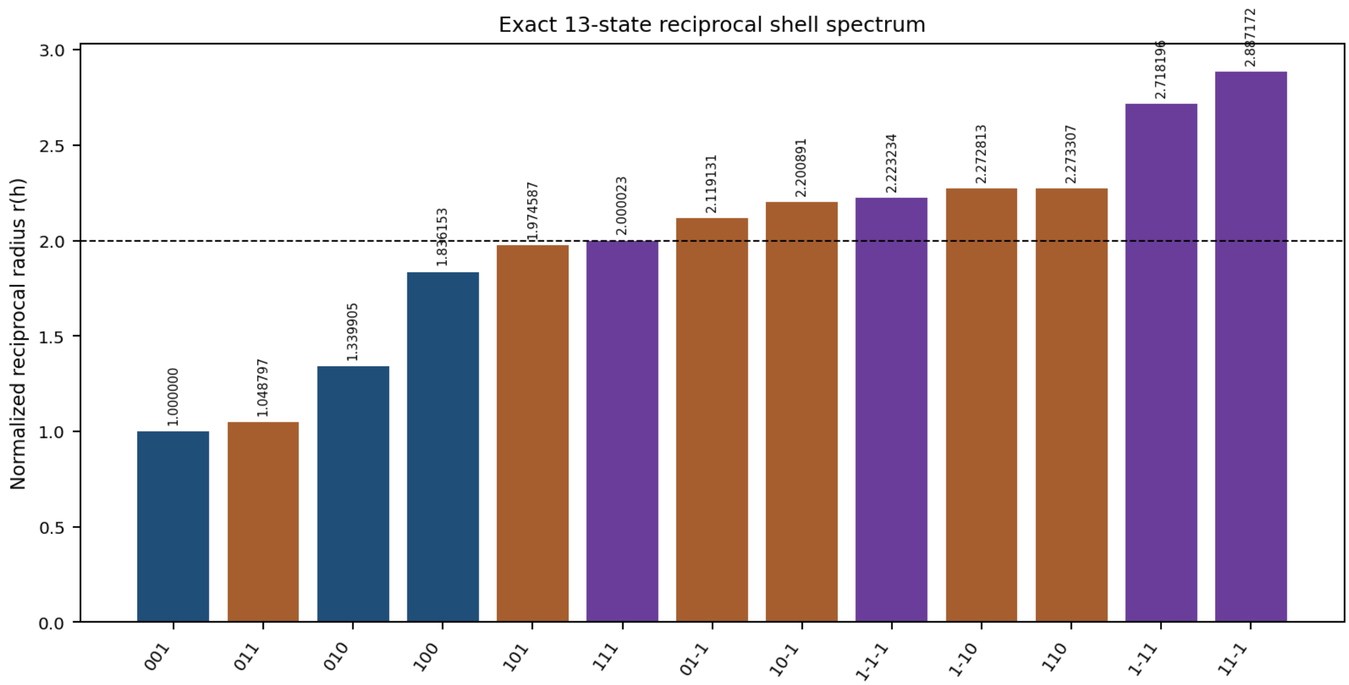

Figure 3. The 13 normalized reciprocal radii, ordered by magnitude and colored by finite quadratic class. The value r(h) is the normalized physical gradient eigenvalue of the ternary Fourier character indexed by h.

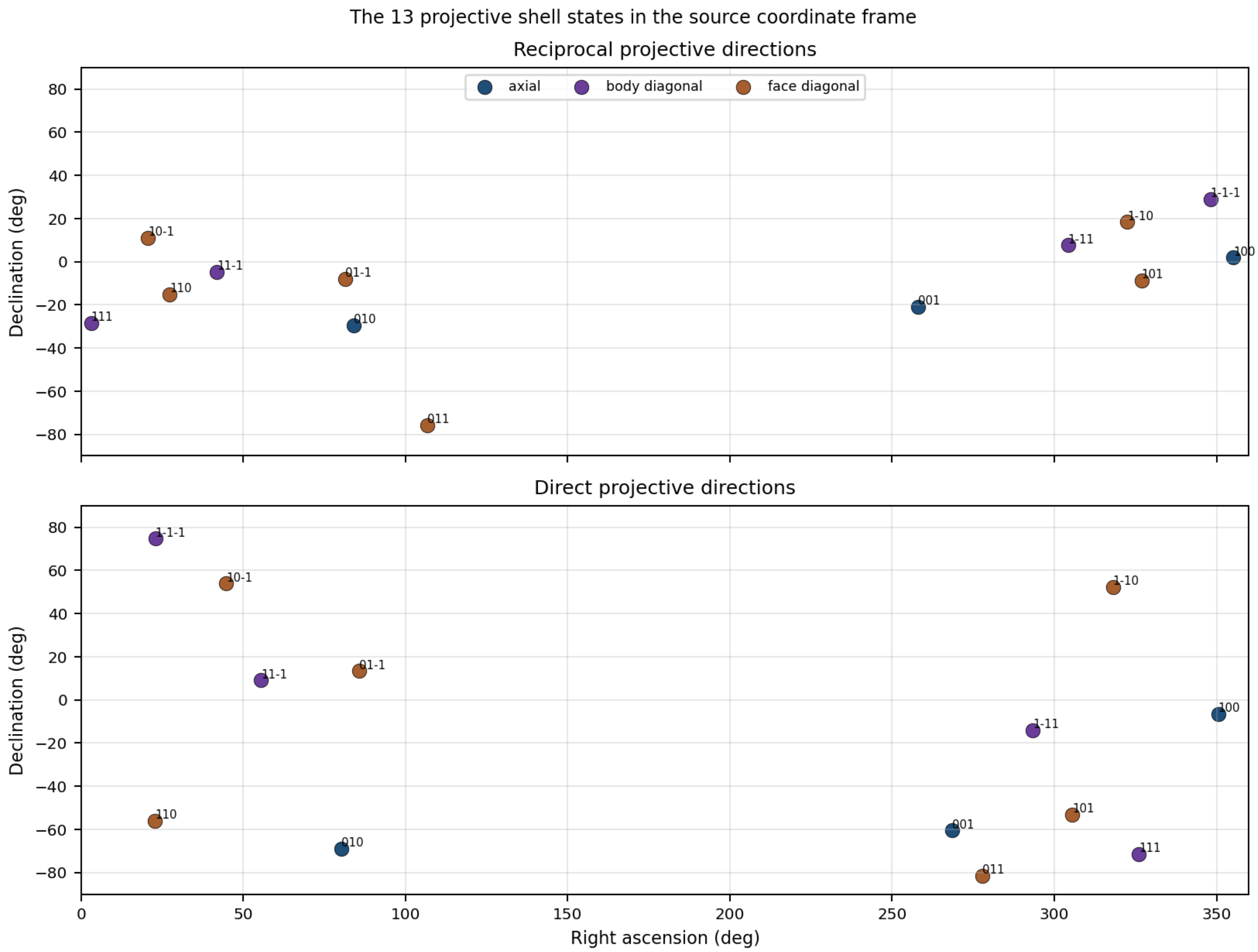

Figure 4. Direct and reciprocal renderings of the same 13 projective addresses in the source celestial coordinate frame. The finite address is invariant; the physical direction changes according to B or B^(-T).

4. Projective incidence as a zero-derivative relation

Translate a Fourier character by t steps along a direct projective direction k. The phase multiplier is omega^(t h dot k). The field is invariant along k precisely when h dot k is zero modulo three. The incidence rule of PG(2,3) is therefore the exact zero-directional-derivative rule for Cell3 Fourier modes.

Each reciprocal mode has four direct directions along which its phase does not change. Each direct direction lies in four reciprocal plane families. This is the finite geometric origin of the four-incidence count. Incidence is not a separate combinatorial ornament placed on top of the spectral system; it is the derivative-null structure of that system.

Let N be the 13 x 13 incidence matrix. Its row and column sums are four. The exact identity N N^T = 3 I + J follows because two distinct projective lines meet in one point while a line contains four points. The eigenvalues of N are 4 once, +sqrt(3) six times, and -sqrt(3) six times. These values depend only on PG(2,3), not on the metric B.

```
psi_h(u+t k) = omega^(t h dot k) psi_h(u)
D_k psi_h = 0 <=> h dot k = 0 mod 3
N 1 = 4 1
N N^T = 3 I + J
spectrum(N) = {4^(1), (+sqrt(3))^(6), (-sqrt(3))^(6)}
```

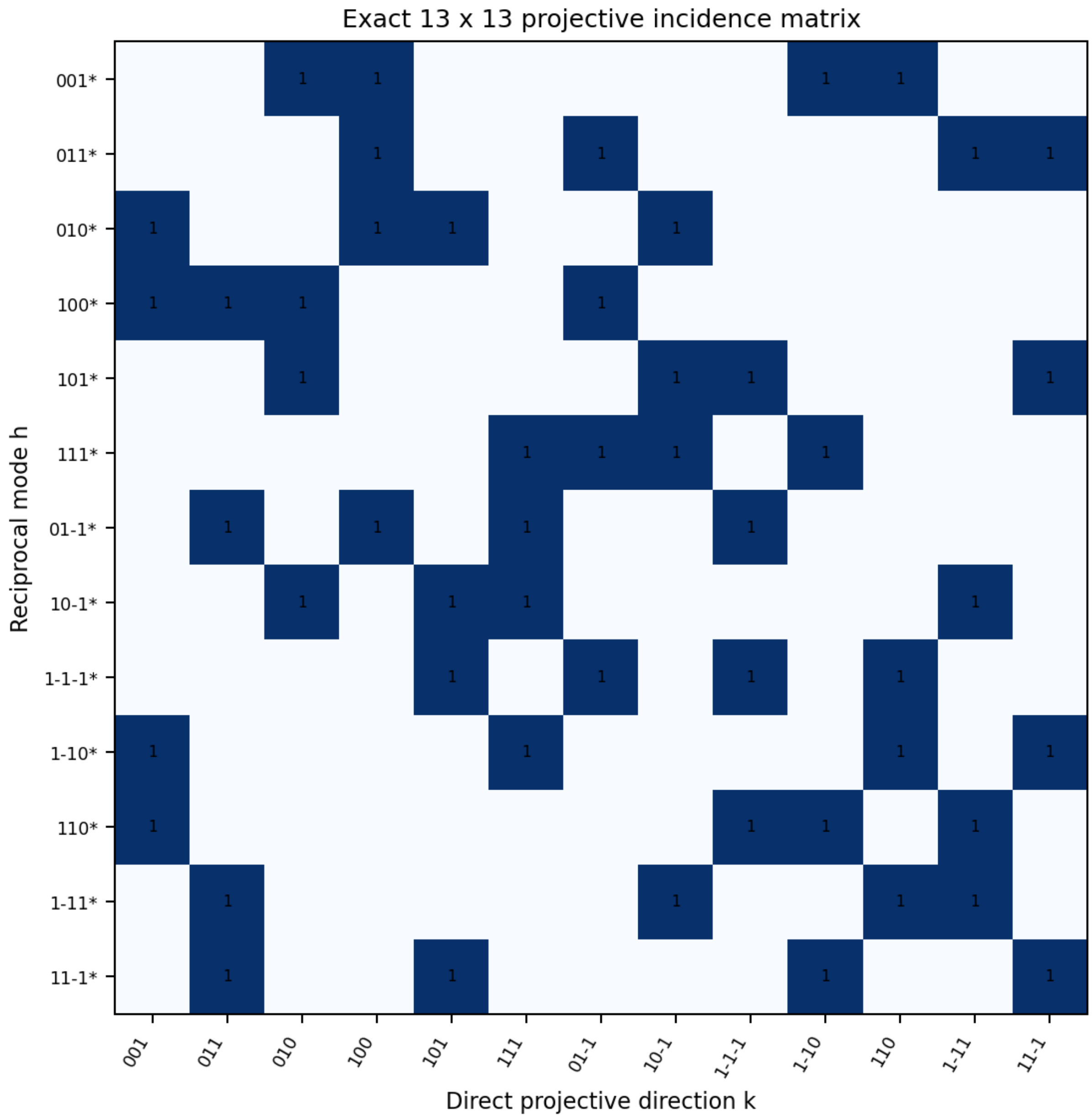

Figure 5. Exact 13 x 13 incidence matrix. A marked cell identifies a reciprocal mode h and a direct direction k satisfying h dot k = 0 mod 3. Every row and column contains four marked cells.

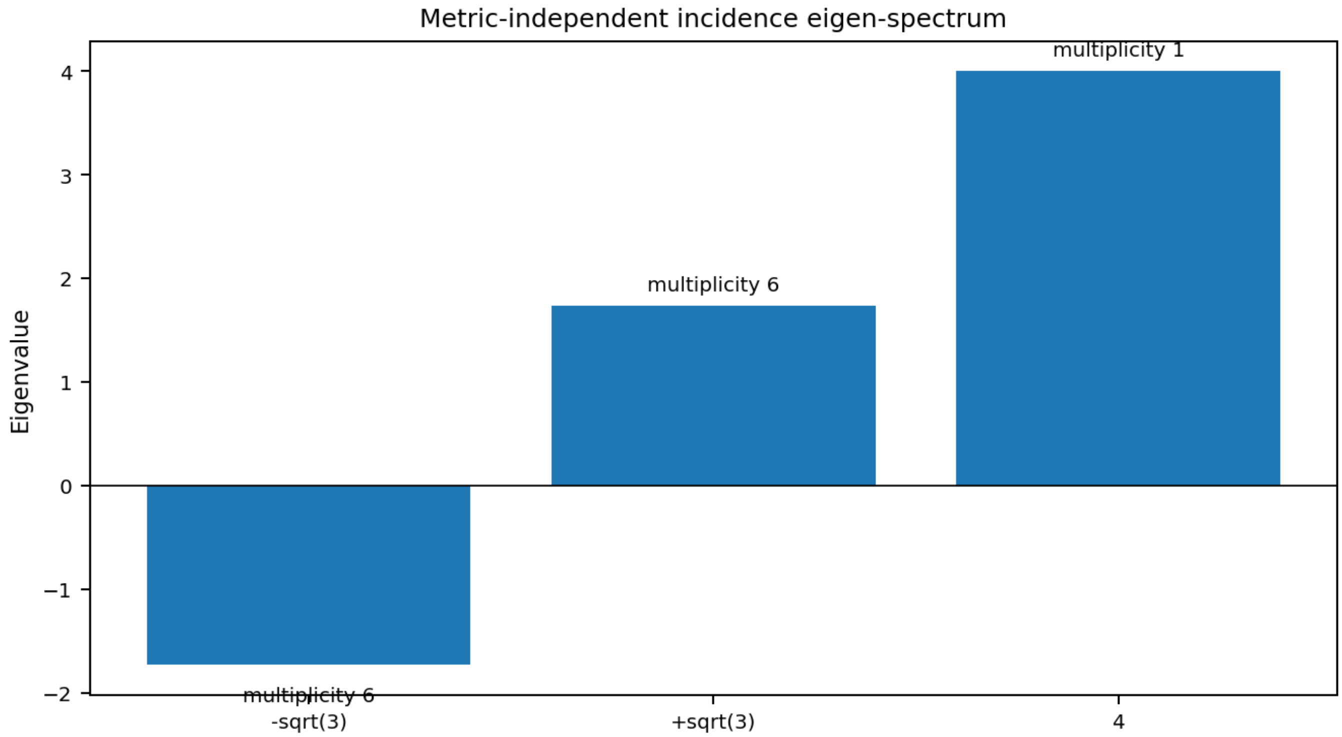

Figure 6. Metric-independent eigen-spectrum of the symmetric point-line incidence matrix. The one-dimensional uniform sector and the two six-dimensional signed sectors are invariant under every nonsingular physical embedding of the finite cell.

4.1 Complete incidence ledger

Reciprocal mode h	Class	Direct zero-derivative directions k	Count
001	axial	010, 100, 1-10, 110	4
011	face diagonal	100, 01-1, 1-11, 11-1	4
010	axial	001, 100, 101, 10-1	4
100	axial	001, 011, 010, 01-1	4
101	face diagonal	010, 10-1, 1-1-1, 11-1	4
111	body diagonal	111, 01-1, 10-1, 1-10	4
01-1	face diagonal	011, 100, 111, 1-1-1	4
10-1	face diagonal	010, 101, 111, 1-11	4
1-1-1	body diagonal	101, 01-1, 1-1-1, 110	4
1-10	face diagonal	001, 111, 110, 11-1	4
110	face diagonal	001, 1-1-1, 1-10, 1-11	4
1-11	body diagonal	011, 10-1, 110, 1-11	4
11-1	body diagonal	011, 101, 1-10, 11-1	4

5. Seven sheets as an A1 plus A2 lattice with ternary gluing

The source coordinates lambda, xi1, and xi2 are exact integer linear combinations of the Lucas address. lambda = i + j + k measures displacement along the 111 address direction. xi1 = i - j and xi2 = j - k measure differences within planes of constant lambda. The transformation has determinant three, so it is not a unimodular relabeling of Z^3. Its image is an index-three sublattice.

Solving for i, j, and k introduces division by three. Integer recovery is possible only when lambda - xi1 + xi2 is zero modulo three. This congruence is the gluing law. It states that the sheet coordinate and the tangential A2 coordinates cannot be selected independently. Each sheet occupies one of three tangential cosets.

The Euclidean address norm decomposes into an A1 longitudinal term and the hexagonal A2 tangential norm. The A2 form xi1^2 + xi1 xi2 + xi2^2 generates the familiar sixfold geometry of a triangular lattice. Cell3 is therefore an A1 plus A2 architecture with index-three gluing, physically sheared by the oblique matrix B.

```
[lambda, xi1, xi2]^T = [[1,1,1],[1,-1,0],[0,1,-1]] [i,j,k]^T
det = 3
i = (lambda+2 xi1+xi2)/3
j = (lambda-xi1+xi2)/3
k = (lambda-xi1-2 xi2)/3
lambda-xi1+xi2 = 0 mod 3
i^2+j^2+k^2 = lambda^2/3 + (2/3)(xi1^2+xi1 xi2+xi2^2)
```

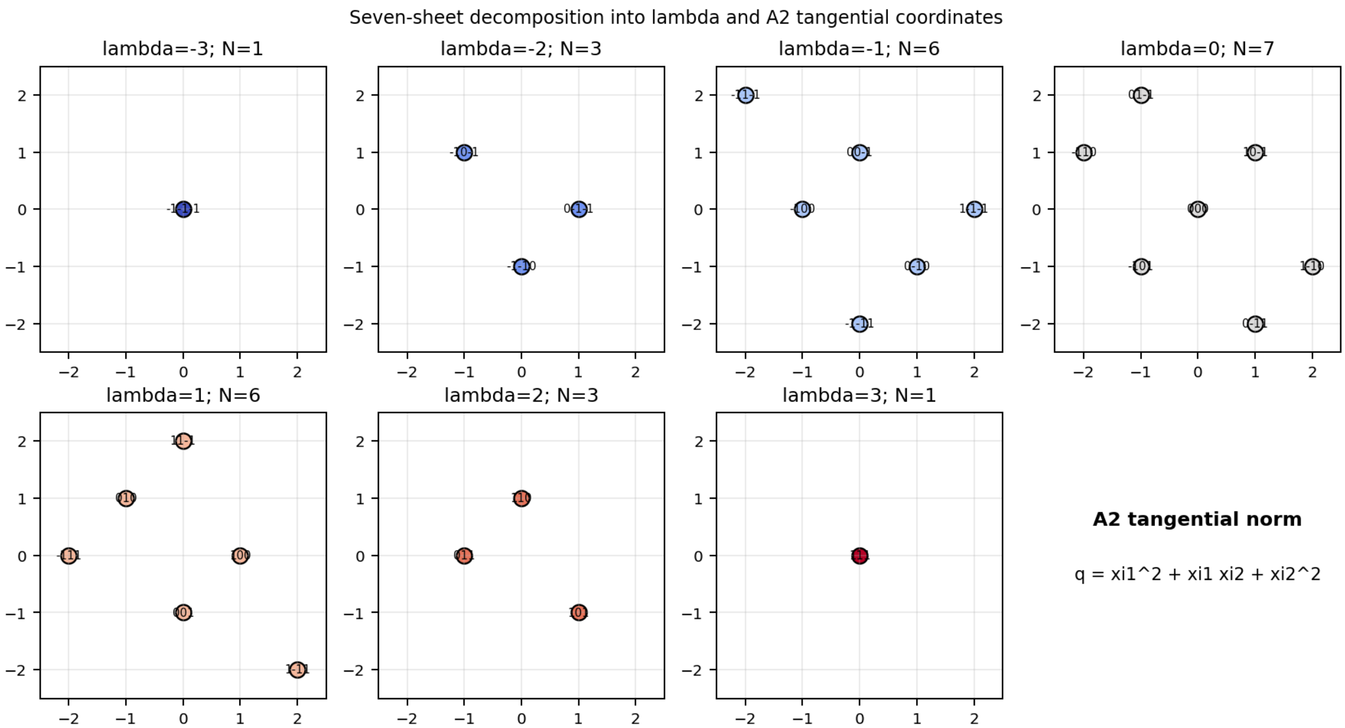

Figure 7. Every Lucas node expressed in the tangential A2 coordinates for each of the seven lambda sheets. The point labels are ternary addresses. Adjacent sheets occupy different congruence cosets, demonstrating the index-three gluing law.

5.1 Exact sheet multiplicities

The coefficient of t^lambda in (t^-1 + 1 + t)^3 counts the number of ternary addresses with coordinate sum lambda. The seven coefficients are 1, 3, 6, 7, 6, 3, and 1. This distribution is forced by the 3 x 3 x 3 address cube.

Reducing lambda modulo three groups the 27 nodes into three phase classes of nine nodes each. These are the three affine planes of the 111 Fourier mode. The visible seven-sheet stack is therefore an integer lift of a three-phase system.

```
(t^-1 + 1 + t)^3 = t^-3 + 3 t^-2 + 6 t^-1 + 7 + 6 t + 3 t^2 + t^3
Sheet populations = 1, 3, 6, 7, 6, 3, 1
Counts by lambda mod 3 = 9, 9, 9
```

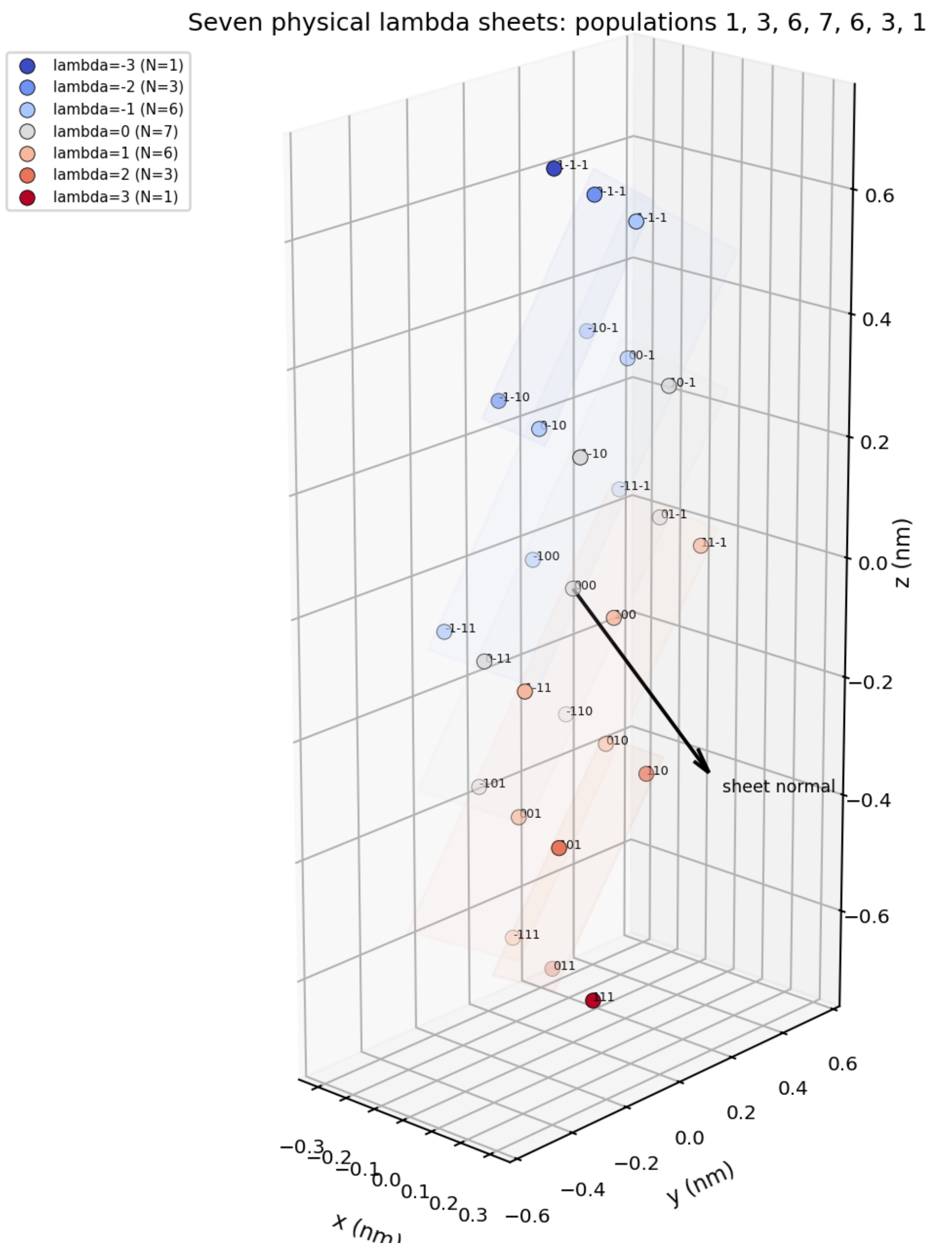

Figure 8. Dimensionally correct seven-sheet stack in physical space. The nodes are colored by lambda. Interior plane patches show the orientation of representative sheets. The black arrow is the reciprocal sheet normal B^(-T)(1,1,1).

6. Binary polarity, ternary phase, and the Z6 registry

The source polarity is chi = (-1)^lambda. Consequently lambda mod 2 determines the binary sign state, while lambda mod 3 determines the ternary Fourier phase class. The pair of residues is equivalent to one residue modulo six because two and three are coprime.

The seven consecutive values lambda = -3 through +3 visit all six binary-ternary registry states and return to the starting residue at the seventh layer. The seventh sheet is therefore the closure display of a six-transition cycle. This explains why a seven-sheet representation naturally accompanies a Z6 registry even though only six distinct phase-polarity combinations exist.

This result unifies sheet position, parity, and phase. They are not three independent labels. They are different quotient renderings of one integer lift lambda.

```
chi = (-1)^lambda
phase = lambda mod 3
polarity residue = lambda mod 2
```

```
(lambda mod 2, lambda mod 3) isomorphic to lambda mod 6
```

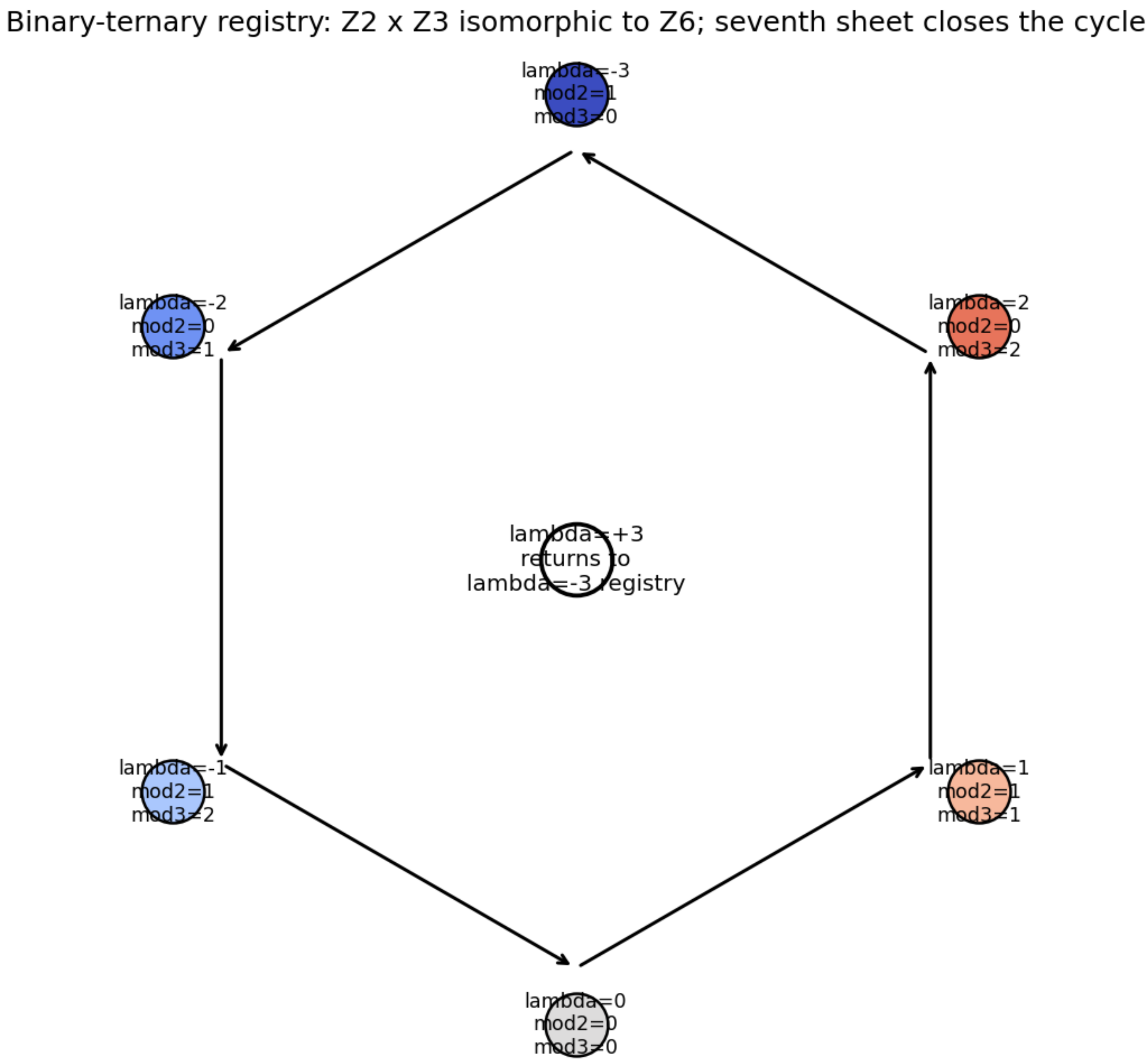

Figure 9. The six binary-ternary registry states and the seventh-sheet return. Each outer node gives lambda modulo two and modulo three. The center identifies the closing +3 sheet, which has the same combined registry as -3.

Hidden axiom 3. Seven-sheet memory is the integer lift of one native Z6 registry variable, not an independent seven-valued primitive.

7. Direct-reciprocal duality as exterior algebra

The cofactor matrix of B has columns b cross c, c cross a, and a cross b. These are oriented face bivectors: each is normal to one primitive face and has magnitude equal to that face area. The exact matrix identity Cof(B) = det(B) B^(-T) connects these bivectors to the reciprocal basis.

For a projective address h, the reciprocal vector B^(-T) h is a determinant-scaled coherent sum of the three face bivectors. The 13 shell radii are therefore the normalized magnitudes of all projectively distinct sums and differences of the three primitive face bivectors. Axial states isolate one face area at a time; diagonal states encode interference between face bivectors.

The Mass Bridge ratios follow immediately. With the 001 state as normalization, r(010) = A_ac/A_ab and r(100) = A_bc/A_ab. The remaining diagonal radii contain the pairwise bivector dot products and complete the normalized reciprocal metric.

```
Cof(B) = [b x c, c x a, a x b] = det(B) B^(-T)
B^(-T) h = [h1(b x c)+h2(c x a)+h3(a x b)] / det(B)
r(h) = |h1(b x c)+h2(c x a)+h3(a x b)| / |a x b|
```

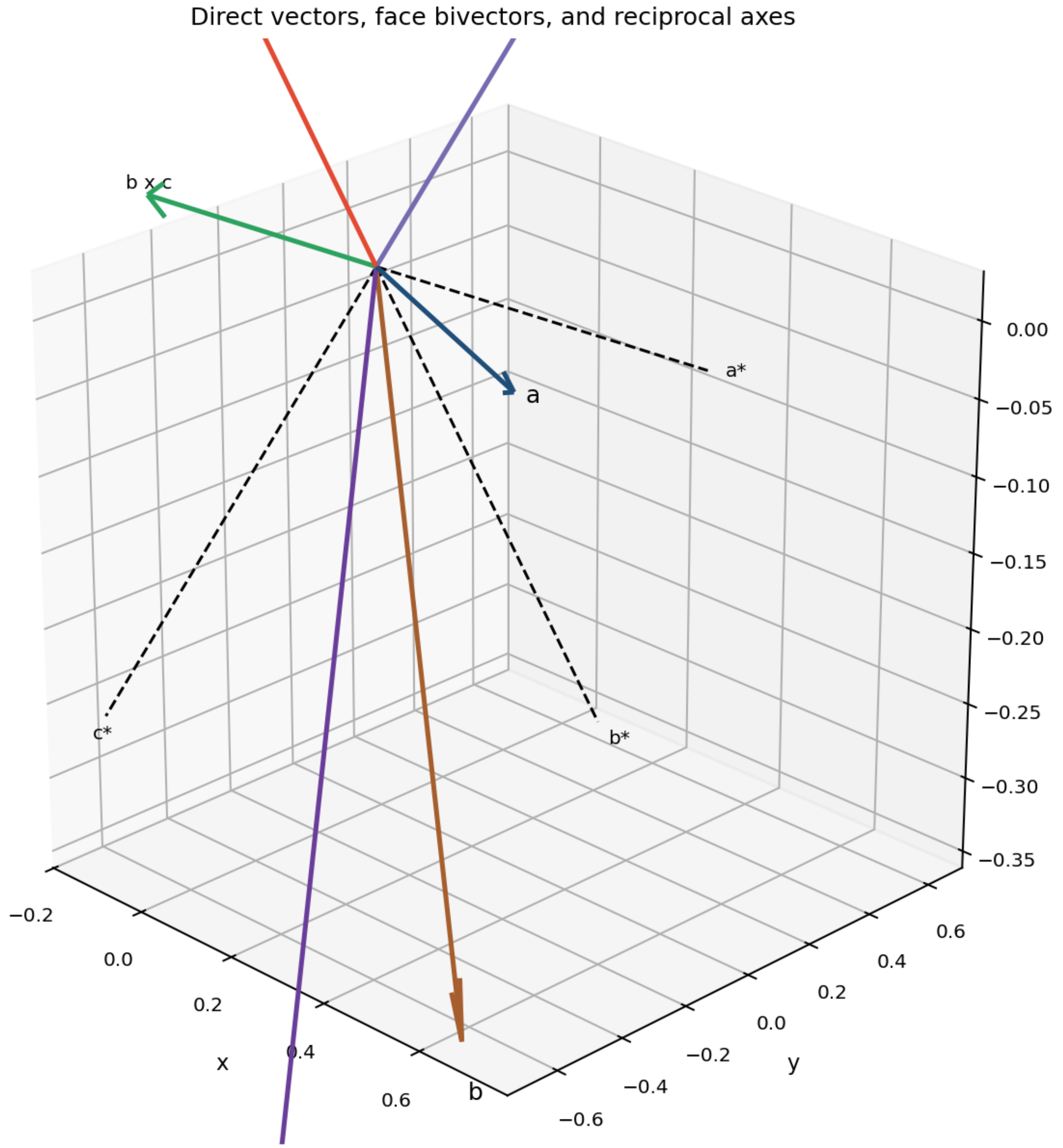

Figure 10. Direct basis vectors, oriented face bivectors, and reciprocal axes. The reciprocal directions are parallel to the cofactor columns and are therefore exterior-grade-2 renderings of the direct cell.

Face area	Value (nm^2)	Axial state	Normalized radius
A_ab	0.194869165469900	001	1.000000000000
A_ac	0.261106235799531	010	1.339905341976
A_bc	0.357809539128562	100	1.836152673337

The exact ratios are A_ac/A_ab = 1.339905341976038, A_bc/A_ab = 1.836152673337283, and A_bc/A_ac = 1.370359991721057. Their multiplicative closure error is zero in the source record.

7.1 Exterior singular values and grade reciprocity

If the singular values of B are s1, s2, and s3, the singular values of its grade-2 exterior map are s1 s2, s1 s3, and s2 s3. Each is the volume divided by the omitted grade-1 singular value. Strong extension in one direct direction therefore corresponds to weak reciprocal response in its dual direction, and vice versa.

```
singular(Lambda^2 B) = {s1 s2, s1 s3, s2 s3}
= |det B| {1/s3, 1/s2, 1/s1}
```

8. Seven exact closure-null identities for the 13 shell radii

A symmetric 3 x 3 reciprocal metric has six independent entries. Normalizing one radial scale removes one degree of freedom, leaving five independent dimensionless parameters. The 13 squared projective radii therefore cannot vary independently. They contain seven redundant degrees of freedom, which appear as seven exact closure-null identities.

Three closures pair the plus and minus face diagonals. Adding R_(ei+ej) and R_(ei-ej) cancels the off-diagonal metric term, leaving twice the two axial squared radii. Four additional closures predict every body-diagonal squared radius from the axial values and three signed face-diagonal differences. Algebraically, the body-diagonal quartet is a four-component Hadamard transform of the metric trace and its three off-diagonal couplings.

These identities provide a stringent reconstruction protocol. Five independent normalized observations determine the entire reciprocal metric and all remaining shell radii. Fitting all 13 radii independently discards the most informative part of the architecture: its exact redundancy.

```
R_(ei+ej) + R_(ei-ej) = 2(R_ei + R_ej)

R_(1,s,t) = R_100 + R_010 + R_001
+ (s/2)(R_110-R_1-10)
+ (t/2)(R_101-R_10-1)
+ (s t/2)(R_011-R_01-1), s,t in {+1,-1}
```

Closure	Expression	Residual
C_ab	R(1, 1, 0)+R(1, -1, 0)-2(R(1, 0, 0)+R(0, 1, 0))	1.7763568394002505e-15
C_ac	R(1, 0, 1)+R(1, 0, -1)-2(R(1, 0, 0)+R(0, 0, 1))	-1.7763568394002505e-15
C_bc	R(0, 1, 1)+R(0, 1, -1)-2(R(0, 1, 0)+R(0, 0, 1))	8.8817841970012523e-16
C_(1, 1, 1)	R(1, 1, 1) - Hadamard prediction	0.0000000000000000e+00
C_(1, 1, -1)	R(1, 1, -1) - Hadamard prediction	-1.7763568394002505e-15
C_(1, -1, 1)	R(1, -1, 1) - Hadamard prediction	8.8817841970012523e-16
C_(1, -1, -1)	R(1, -1, -1) - Hadamard prediction	0.0000000000000000e+00

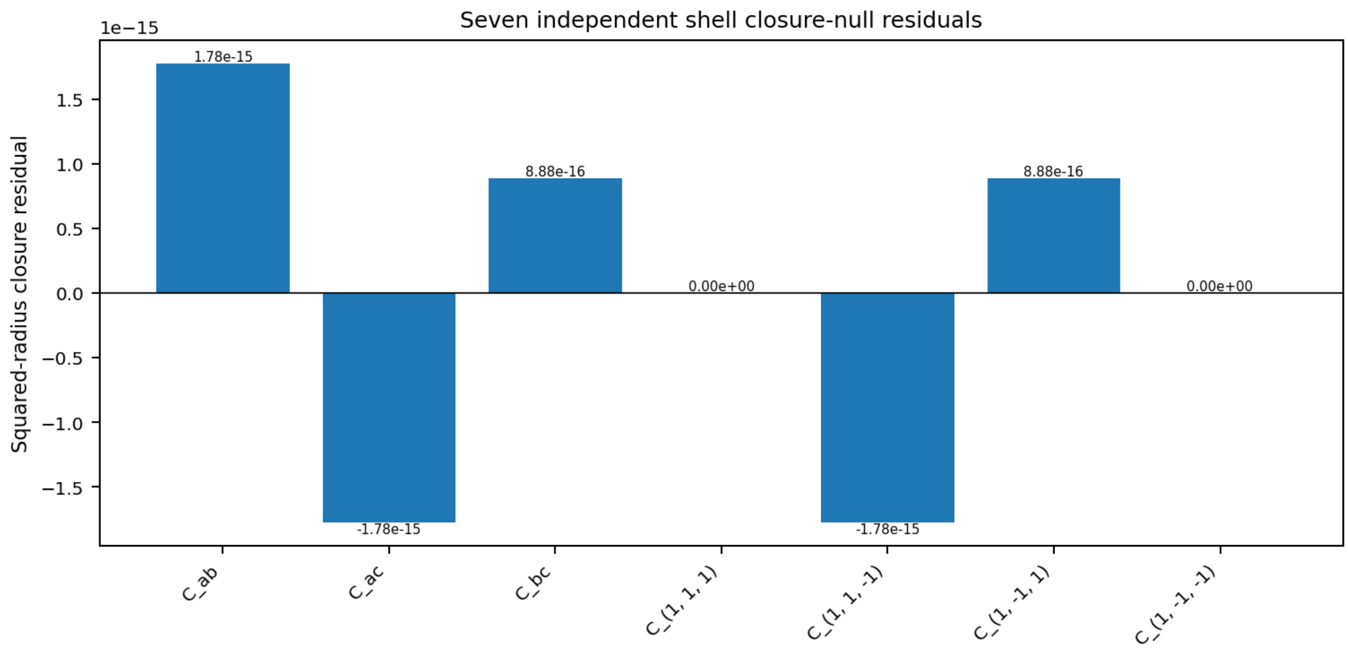

Figure 11. Computed residuals of all seven independent shell closures. Values are at floating-point reconstruction scale; no closure requires an additional fitted parameter.

8.1 Hidden conditional orthogonality

The normalized reciprocal metric has an exceptionally small a-star/b-star cross term. Its normalized cosine is only 2.284129 x 10^-4, corresponding to a reciprocal angle of 89.9869129 degrees. In direct geometry, the residual components of a and b after removing their projections through c meet at the complementary angle 90.0130871 degrees.

The result is a conditional orthogonality. Most of the direct a-b coupling is transmitted through the third basis direction. Setting the reciprocal cross term exactly to zero would make the 110 and 1-10 shell radii degenerate. The present departure is a sharply localized refinement parameter rather than a broad deformation of the metric.

```
cos(a* , b*) = 2.284129030852e-04
angle(a* , b*) = 89.986912905 deg
conditional direct angle = 90.013087095 deg
```

8.2 The 111 binary-ternary span lock

The body-diagonal radius r(111) equals 2.000023101807, only 11.5508 parts per million above two. The exact value two has a direct registry interpretation. Adjacent physical phase planes of mode h are separated by a distance inversely proportional to |B^(-T) h|. Six 111 sheet intervals span the same distance as three axial 001 intervals exactly when r(111)=2.

Thus the near-integer 111 radius couples the six-transition Z6 registry to a three-interval axial Cell3 span. This is a binary-ternary metric lock: six registry transitions close against three axial intervals through the factor two.

```
r(111) = 2.000023101807
[r(111)/2 - 1] x 10^6 = 11.550903 ppm
6 d_111 = 3 d_001 <=> r(111)=2
```

9. Physical shear of the seven-sheet stack

Address planes of constant lambda have physical normal B^(-T)(1,1,1). Their centroids advance along B(1,1,1). In an orthogonal isotropic cell these two directions would coincide. In the authoritative Cell3 they differ by 47.507119 degrees.

Each layer transition therefore contains both normal advance and tangential displacement. The tangential displacement is 1.09158 times the normal advance. The visible stack is a sheared foliation rather than a set of normally translated copies.

This shear supplies a concrete geometric mechanism for registry memory. Moving from one sheet to the next necessarily changes the tangential A2 coset as well as the longitudinal coordinate. A closed excursion through sheet space can therefore retain a tangential phase displacement even when its net longitudinal change is zero.

```
normal direction = unit[B^(-T)(1,1,1)]
centroid advance = unit[B(1,1,1)]
projective angle = 47.507119364 deg
tangential/normal advance = tan(angle) = 1.091580778
```

Geometric consequence. Sheet registry and tangential holonomy are inseparable because the physical stack is oblique to its own reciprocal normal.

10. Exact circular sections of the phase metric

The phase cost is a positive-definite quadratic form x^T M x. Its three eigenvectors define soft, middle, and stiff axes; its eigenvalues determine the corresponding squared costs. In the eigenbasis, the unit-cost surface is an ellipsoid with semiaxes 1/sqrt(lambda1), 1/sqrt(lambda2), and 1/sqrt(lambda3).

A triaxial ellipsoid has two central planes whose intersections are exact circles. For the phase tensor, define rho = (lambda2-lambda1)/(lambda3-lambda1), a = sqrt(rho), and b = sqrt(1-rho). The vectors w_plus = b e1 + a e3 and w_minus = b e1 - a e3, together with e2, span the two circular-section planes. On either plane the restricted quadratic form is exactly lambda2 times the identity.

Every unit direction in either plane has the same phase cost. These are not approximate low-modulation paths obtained from a numerical sweep. They are exact zero-modulation great circles determined by the eigenvalues. Their two normals form the reported projective angle of 35.450129 degrees.

```
rho = (lambda2-lambda1)/(lambda3-lambda1)
a = sqrt(rho), b = sqrt(1-rho)
w+ = b e1 + a e3
w- = b e1 - a e3
P+ = span(e2,w+), P- = span(e2,w-)
M restricted to P+ = lambda2 I
M restricted to P- = lambda2 I
```

Quantity	Value	Cost sqrt(lambda)
lambda1	1.429328753487483	1.195545379100051
lambda2	2.112545026447479	1.453459674861150
lambda3	8.800336434665038	2.966536100347514
rho	0.092689670464547	-
a	0.304449783157334	-
b	0.952528387784560	-
plane angle	35.450129228842 deg	-

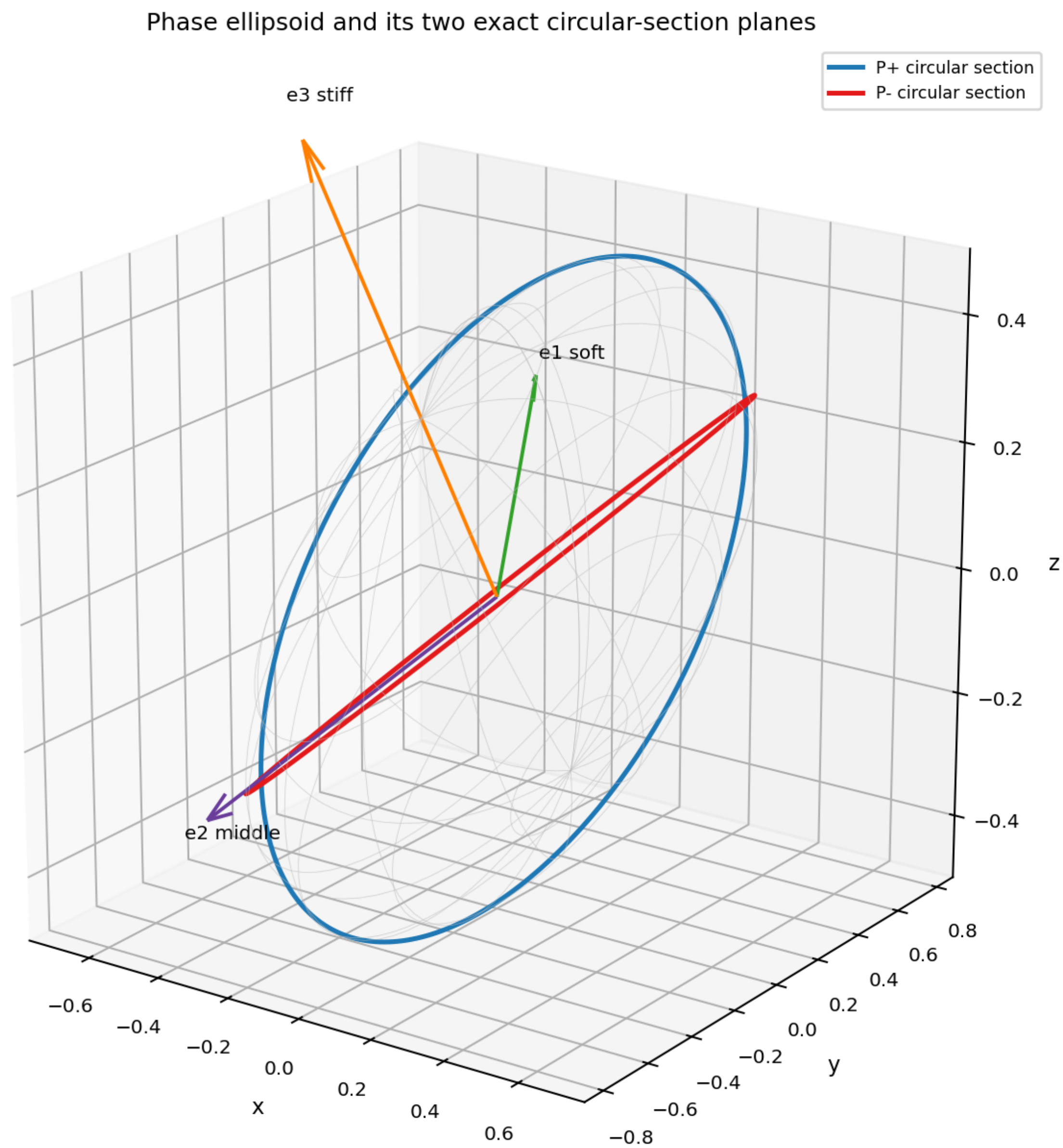

Figure 12. Phase ellipsoid generated from the exact tensor eigenvalues and eigenvectors. The blue and red curves are the two exact circular central sections. Each point on either circle has quadratic cost lambda2.

10.1 Six-node intersection skeleton

The two circular planes meet along the middle eigenaxis e2. Each plane also contains one additional distinguished direction w_plus or w_minus. Antipodal completion produces six nodes: plus and minus e2, plus and minus w_plus, and plus and minus w_minus. This is the complete phase-metric intersection skeleton.

The skeleton provides a finite directional scaffold for dynamical branch conditions. When an external constraint selects one circular plane, the admissible zero-modulation path is continuous within that plane, but its intersections with other exact structures reduce to the six antipodal nodes.

Node	x	y	z	RA deg	Dec deg	Antipode RA	Antipode Dec
e2	-0.828553895965	-0.256832305478	-0.497529504999	197.222212119	-29.836687522	17.222212119	29.836687522
w_plus	-0.505971380457	0.723974494495	0.468885800041	124.948934124	27.961995685	304.948934124	-27.961995685
w_minus	-0.273107431263	0.961095867883	-0.041316627733	105.863163197	-2.367942424	285.863163197	2.367942424

10.2 Closed analytic modulation amplitude

For a plane normal n, restrict M to the great-circle plane n-perpendicular. The two eigenvalues of that restriction determine the maximum and minimum phase cost around the circle. Their half-difference is the modulation half-amplitude.

The restriction trace is tr(M)-n^T M n. Its determinant is det(M) n^T M^(-1) n. Substituting these two invariants into the quadratic eigenvalue formula gives a closed expression for the amplitude. The two circular-section normals are exactly the projective roots where the radicand vanishes.

```
A(n) = 0.5 sqrt{ [tr(M)-n^T M n]^2 - 4 det(M) n^T M^(-1) n }
```

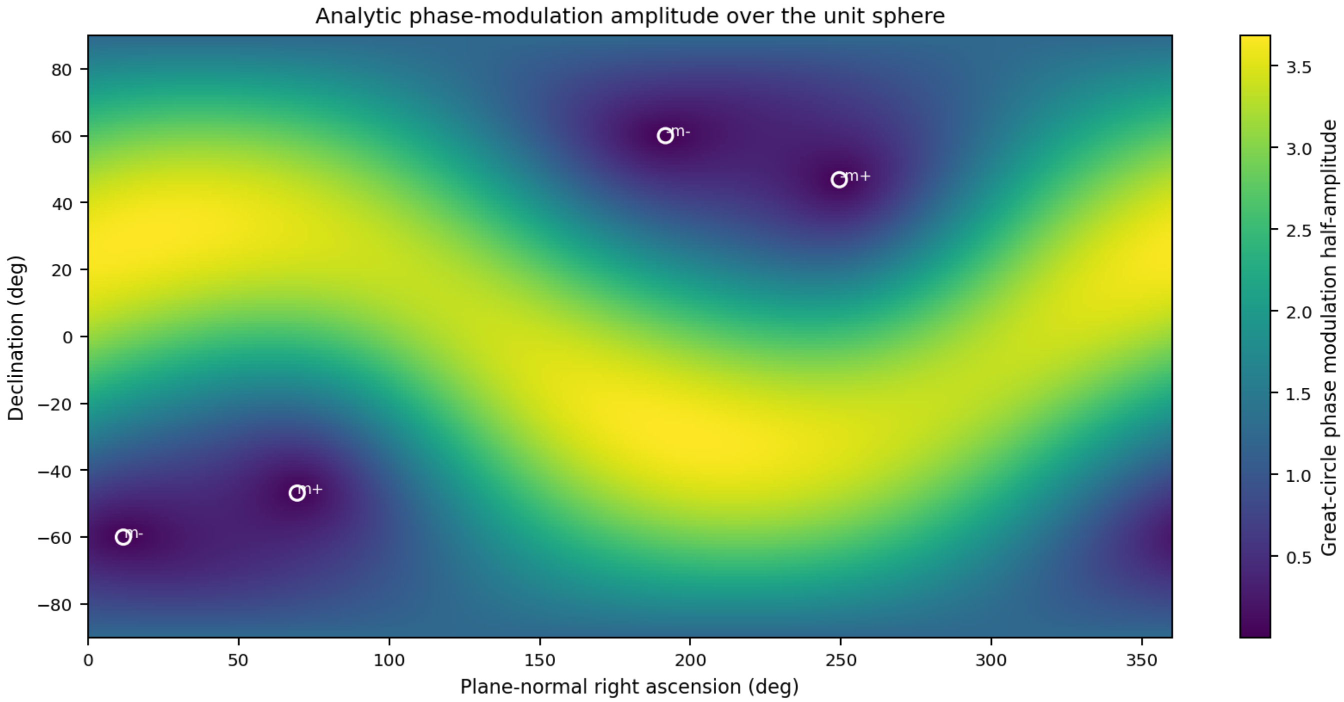

Figure 13. Analytic phase-modulation half-amplitude for every plane normal on the unit sphere. The four marked normals are the antipodal pairings of the two exact circular-section planes and lie at zero modulation.

11. The rank-two binary-ternary scale lattice

The basis law B_(p,n) = 3^p 2^(-n) B and the ratio law r = 2^sigma (2/3)^nu are two coordinate systems on the same multiplicative lattice. Matching powers of two and three gives p = -nu and n = -(sigma+nu). The inverse is nu = -p and sigma = p-n.

The integer transformation has determinant one, so it belongs to GL(2,Z). No scale states are lost or added by changing coordinates. The two descriptions are unimodular bases of the rank-two valuation lattice generated by the primes two and three.

This scale geometry parallels the local registry geometry. Ternary recursion is carried by the power of three; binary branching and parity are carried by the power of two. The architecture is therefore binary-ternary in both internal phase registry and external scale recursion.

```
B_(p,n) = 3^p 2^(-n) B
r = 2^sigma (2/3)^nu
p = -nu
n = -(sigma+nu)
nu = -p
sigma = p-n
determinant of address change = 1
```

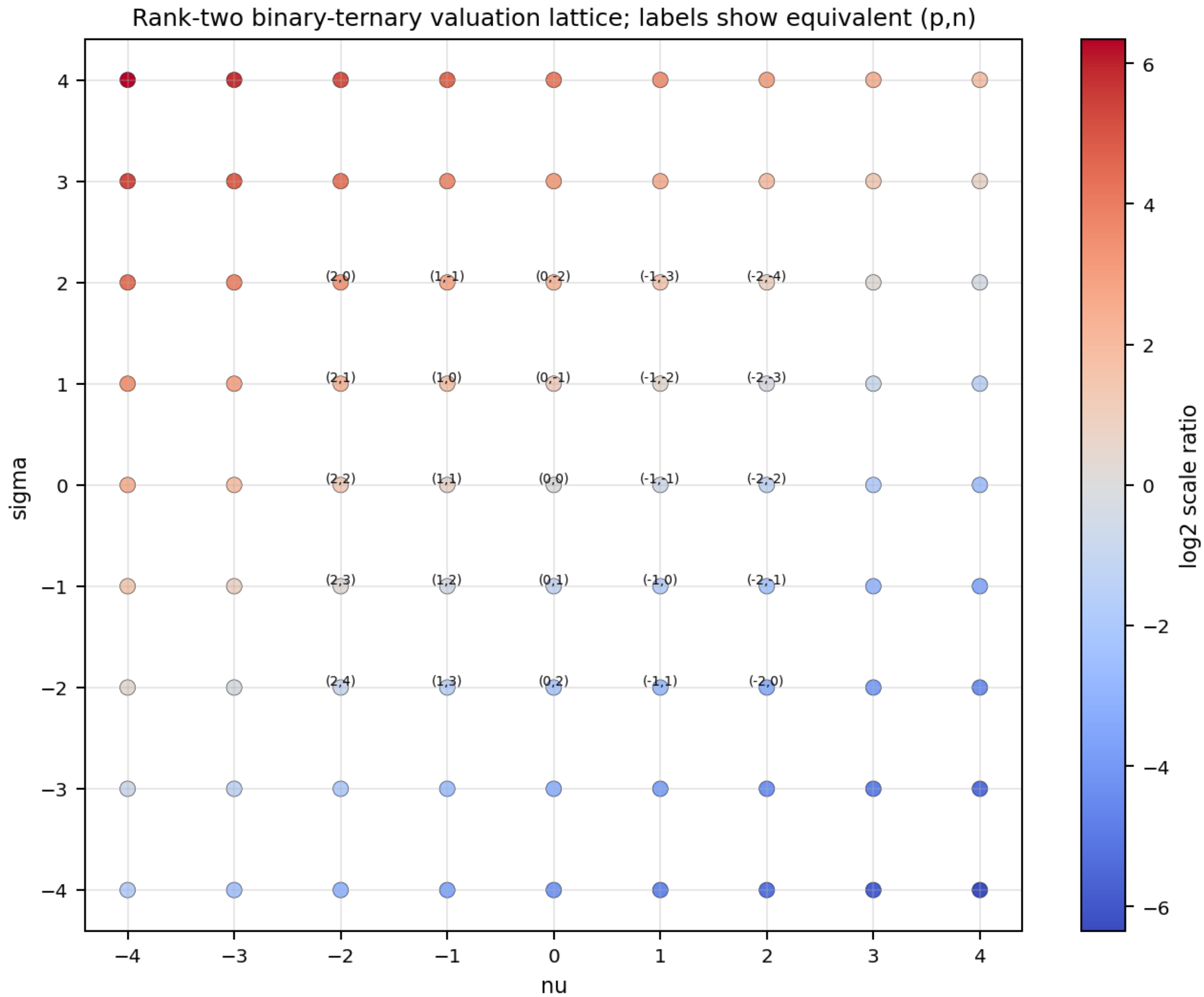

Figure 14. Integer scale-address lattice. Each point is labeled by the equivalent (p,n) address for the same (nu,sigma) state. Color encodes the logarithmic scale ratio.

12. Orbital dimensional collapse and closure-null structure

The solar-system ledger contains seven positive observables: mass, orbital period, semimajor axis, mean orbital speed, centripetal acceleration, total orbital angular momentum, and mean orbital kinetic energy. Taking logarithms converts their multiplicative relations into affine linear relations.

Mean speed is determined by semimajor axis and period. Centripetal acceleration is determined by semimajor axis and period. Kinetic energy is determined by mass and speed. Angular momentum is determined by mass, semimajor axis, speed, and eccentricity. The seven-variable record therefore has only four primitive coordinates: log mass, log semimajor axis, log period, and eccentricity.

The reported PCA confirms this algebraic reduction. The first three components explain 99.99999517 percent of the variance. A fourth component carries only the small eccentricity variation; the remaining directions are numerical closure-null dimensions. The PCA is therefore not merely an empirical compression. It is the spectral signature of exact kinematic identities.

```
log P + log v - log A = log(2*pi)
2 log P + log a_c - log A = log(4*pi^2)
log m + 2 log v - log K = log 2
log L = log m + log A + log v + 0.5 log(1-e^2)
```

Component	Eigenvalue	Variance fraction	Percent
PC1	4.925714828640811e+00	7.036735469486872e-01	70.367354695%
PC2	1.500617345852168e+00	2.143739065503097e-01	21.437390655%
PC3	5.736674875212692e-01	8.195249821732416e-02	8.195249822%
PC4	3.379857532478015e-07	4.828367903540021e-08	0.000004828%
PC5	3.861987003495528e-16	5.517124290707897e-17	0.000000000%
PC6	2.709046763699435e-16	3.870066805284907e-17	0.000000000%
PC7	-5.471157582340366e-16	-7.815939403343380e-17	-0.000000000%

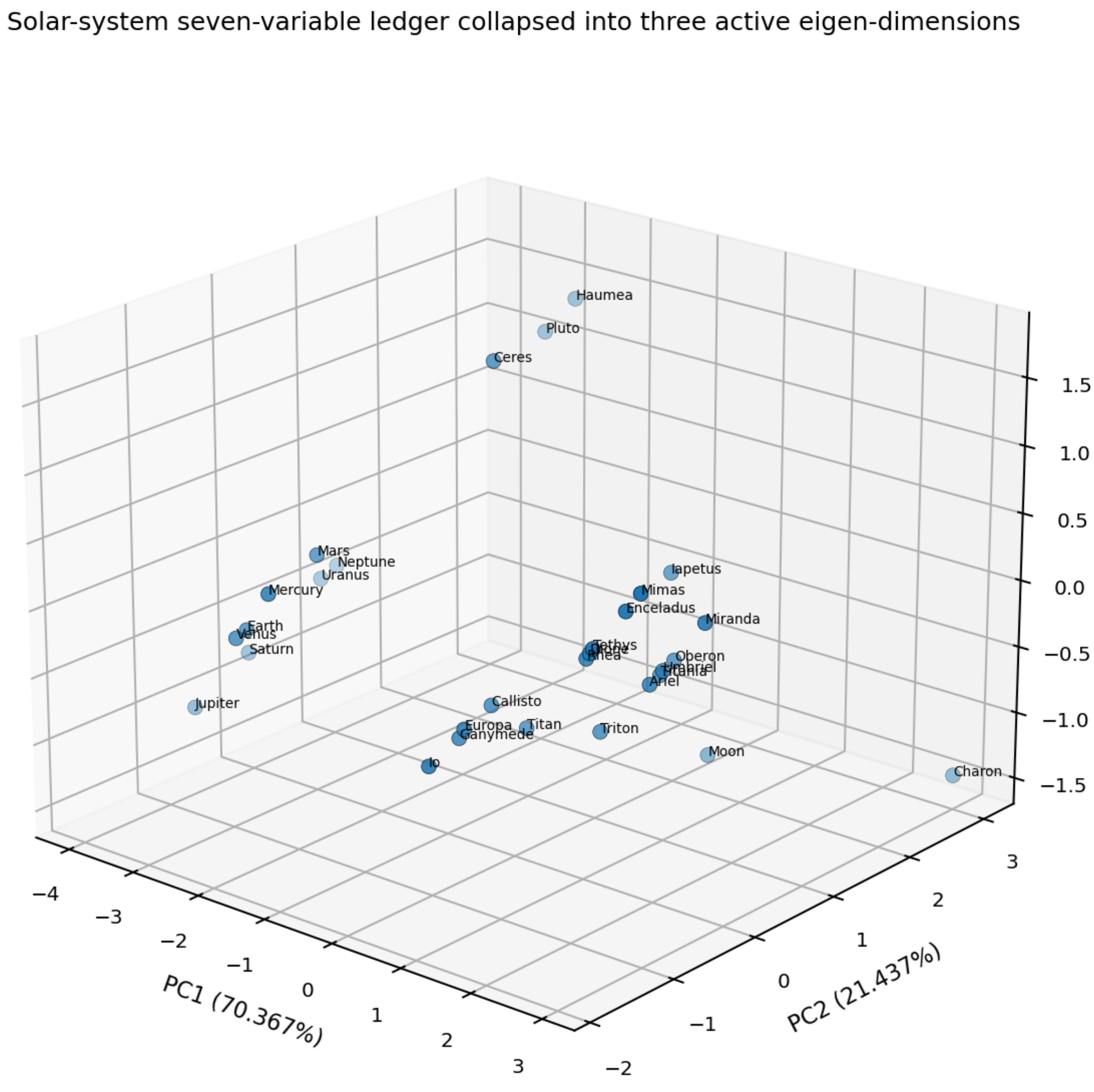

Figure 15. The 30 orbit-bearing bodies projected into the first three principal components of the seven-variable logarithmic ledger. Labels identify each body. These three axes account for essentially all observed variance.

12.1 Primitive-only modular analysis

A modular address should be assigned only to primitive variables. Addresses for speed, acceleration, kinetic energy, and angular momentum must then be derived through the closure equations. Otherwise algebraically dependent quantities are counted as independent confirmations.

Let delta_x denote the residual of a logarithmic variable relative to its assigned modular address. The null relations propagate residuals linearly: delta_v = delta_A - delta_P; delta_ac = delta_A - 2 delta_P; delta_K = delta_m + 2 delta_v; and delta_L = delta_m + delta_A + delta_v plus the eccentricity correction. Any remaining nonclosure after this propagation identifies genuine additional structure.

```
delta_v = delta_A - delta_P
delta_ac = delta_A - 2 delta_P
delta_K = delta_m + 2 delta_v
delta_L = delta_m + delta_A + delta_v + delta_e
```

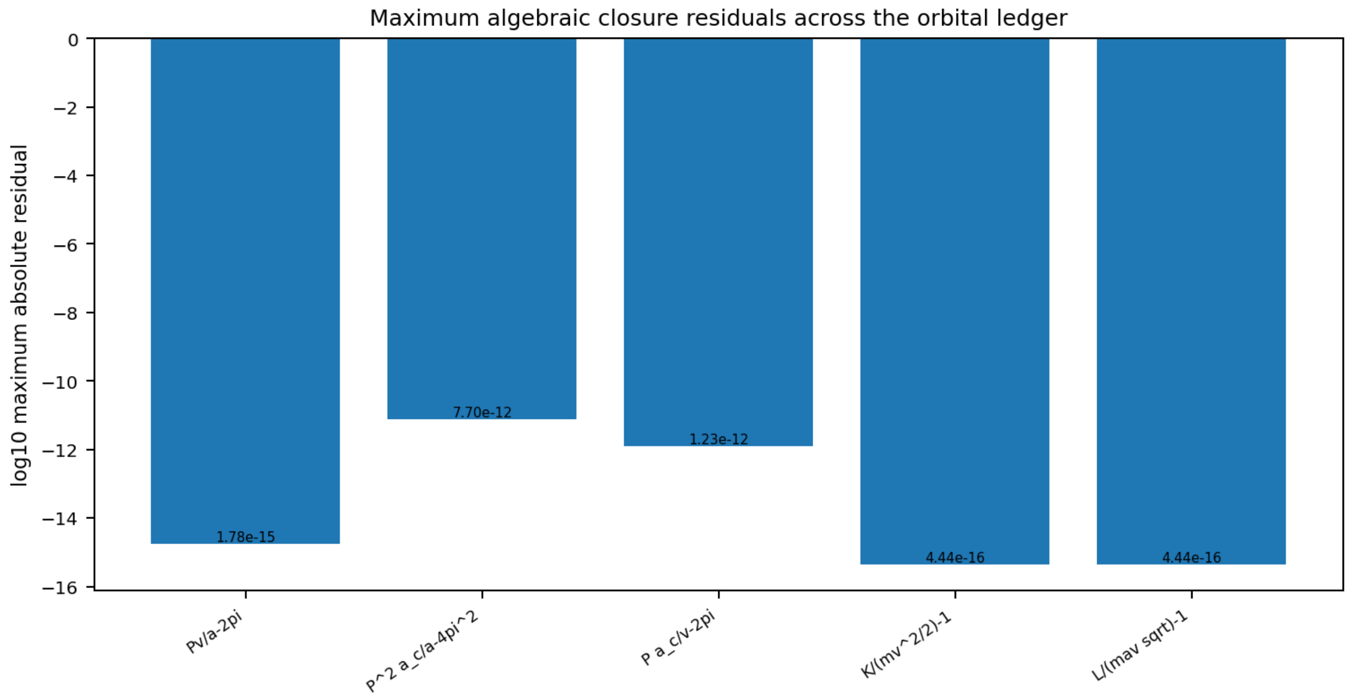

Figure 16. Maximum absolute closure residual for each exact orbital identity over the complete source ledger. Residuals remain at floating-point arithmetic scale.

13. Reduced native state and dependency structure

The source minimal state lists exterior grade g, projective direction [z], scale address p and n, sheet coordinate lambda, A2 coordinates xi1 and xi2, polarity chi, phase registry phi, and occupation mu. The preceding reductions show that several entries are derived within the native Lucas architecture.

The integer address u determines lambda, xi1, xi2, and the projective direction. lambda determines polarity and ternary phase residue. The two scale addresses can be replaced by the binary and ternary valuations v2 and v3. A reduced native state therefore requires exterior grade, one integer lattice address, two scale valuations, phase, and occupation.

This reduction does not discard observable renderings. It separates primitive state variables from functions of those variables. Every expanded coordinate can be reconstructed when required for measurement or display.

```
Source: S = (g,[z],p,n,lambda,xi1,xi2,chi,phi,mu)
Reduced: S_native = (g,u,v3,v2,phi,mu)
Equivalent: S_native = (g,lambda,xi1,xi2,v3,v2,phi,mu)
Constraint: lambda-xi1+xi2 = 0 mod 3
```

14. Unified hidden architecture

The finite, spectral, metric, registry, phase, scale, and orbital structures now form one connected hierarchy. AG(3,3) supplies 27 affine nodes. Antipodal quotienting supplies the 13 projective rays of PG(2,3). The character group of AG(3,3) supplies 13 nonconstant conjugate Fourier sectors indexed by those same rays. The physical basis B and its exterior dual render each sector as a direct direction, reciprocal plane normal, and gradient-response magnitude.

The integer address separates into a longitudinal A1 coordinate and tangential A2 coordinates subject to an index-three glue law. The longitudinal coordinate carries a ternary phase and a binary polarity, which combine into Z6. The seven visible sheets are one closed display window of that registry. At larger and smaller scales, the same primes two and three generate a rank-two valuation lattice.

The phase tensor contributes a continuous quadratic geometry whose two exact circular sections reduce its zero-modulation structure to two planes and six antipodal skeleton nodes. The orbital ledger demonstrates a parallel principle of explanatory compression: multiple measured variables lie on a low-dimensional manifold because they are generated from a smaller primitive state.

The paradigm is therefore neither purely discrete nor purely continuous. A finite ternary projective Fourier substrate carries exact state and incidence relations. Real matrices render that substrate into continuous physical coordinates, anisotropic response, and scale-dependent observables. Closure identities connect the two representations and prevent redundant quantities from becoming independent degrees of freedom.

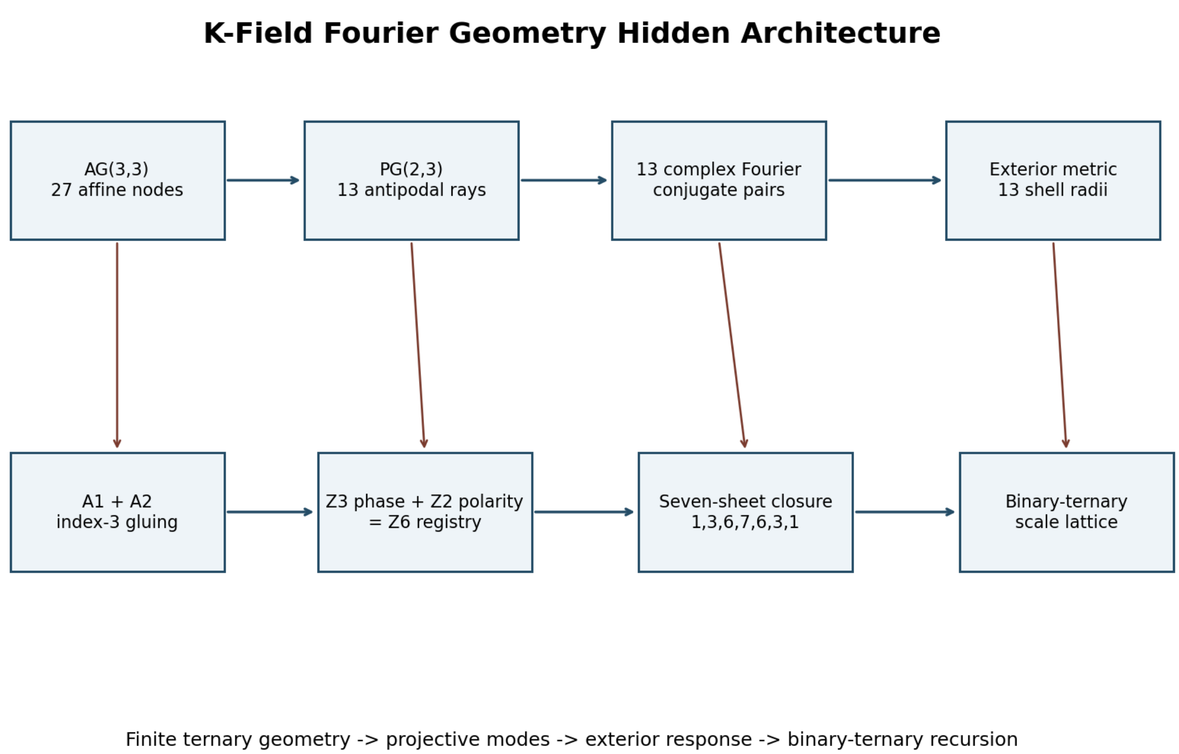

Figure 17. Unified architecture. The upper chain identifies the finite Fourier and exterior-response structure. The lower chain identifies the A1-A2, Z6 registry, seven-sheet closure, and binary-ternary scale recursion. Vertical links show the shared address substrate.

Hidden master axiom. K-Field structure is a ternary affine Fourier geometry, projectivized by antipodal equivalence, physically rendered through an exterior metric, and recursively scaled on a binary-ternary valuation lattice.

15. Worked derivations

15.1 Reconstructing a reciprocal shell radius

Choose h = (1,0,1), the projective state 101. Multiply h by the reciprocal metric and take the quadratic product h^T G^(-1) h. Divide by the 001 axial value G^(-1)_33 and take the square root. The result is r(101) = 1.974587074999.

The same result follows in exterior form by adding the bivectors b cross c and a cross b, taking the magnitude, and normalizing by |a cross b|. Agreement of these routes demonstrates that reciprocal metric response and exterior face composition are the same calculation.

```
h^T G^(-1) h = 10.648820732076091
r(101) = sqrt(10.648820732076091/2.731171274746109) = 1.974587074998623
exterior route = |(b x c)+(a x b)|/|a x b| = 1.974587074998623
```

15.2 Finding zero-derivative directions for one mode

For h = 111, evaluate h dot k modulo three for every projective direction k. The four zero values identify the direct directions along which the 111 Fourier phase is invariant.

```
Zero-derivative directions for h=111: 111, 01-1, 10-1, 1-10
```

15.3 Recovering an integer address from sheet coordinates

Take lambda = 1, xi1 = 1, and xi2 = 0. The gluing congruence is satisfied because 1-1+0=0 mod 3. Substitution into the inverse transform gives i=1, j=0, and k=0, recovering the Lucas address (1,0,0).

```
i=(1+2+0)/3=1
j=(1-1+0)/3=0
k=(1-1-0)/3=0
```

15.4 Constructing a phase circular section

The phase eigenvalue cross-ratio is rho = 0.092689670464547. Its square roots give a = 0.304449783157334 and b = 0.952528387784560. The vector w_plus = b e1 + a e3 is orthogonal to e2 and has unit length. In the basis (e2,w_plus), the restricted tensor differs from lambda2 I by only 1.867e-15 in Frobenius norm.

15.5 Predicting an orbital derived residual

Suppose modular assignments produce primitive residuals delta_A = 0.0010 and delta_P = -0.0005. The speed residual is not independently fitted; it is delta_v = 0.0015. The centripetal-acceleration residual is delta_ac = 0.0020. If delta_m = -0.0002, then delta_K = -0.0002 + 2(0.0015) = 0.0028.

16. New experimental and analytical pathways

- **Finite Fourier tomography.** Transform observed directional or occupancy data into the 13 conjugate character sectors rather than assigning only the nearest projective ray. Mode amplitude, phase, and incidence-null structure distinguish direct occupancy from reciprocal-plane organization.
- **Full affine-plane enumeration.** Use all 39 planes h dot u = r mod 3 and all 117 affine lines as the native discrete geometry for registry, crossing, and holonomy analysis.
- **Absolute-conic testing.** Test the four self-incident body-diagonal states as the primary candidate set for invariant propagation, soft-axis locking, and branch-stable dynamics.
- **Five-parameter metric reconstruction.** Measure a minimal independent set of normalized shell responses, reconstruct G^(-1), and predict the other states through the seven exact closures.
- **Conditional-orthogonality refinement.** Measure the 110 versus 1-10 splitting as a direct probe of the very small reciprocal a-star/b-star coupling.
- **111 span-lock refinement.** Test whether higher-precision geometry converges to r(111)=2, the exact closure between six registry transitions and three axial intervals.
- **Analytic phase-axis search.** Evaluate A(n) directly instead of sampling great circles. This yields exact zero-modulation normals and local sensitivity tensors.
- **Primitive-only orbital MOD analysis.** Assign addresses to mass, semimajor axis, period, and eccentricity only. Derive all dependent variables and use closure residuals as the actual test statistic.
- **Cross-domain projective matching.** Back-project astrophysical plane crossings, spin axes, and mechanical force extrema through B or B^(-1) and test for concentration on low-complexity projective addresses.

- **Registry-holonomy devices.** Construct trajectories that close in lambda but not in the tangential A2 registry, exploiting the measured physical sheet shear.

17. Complete 13-state projective shell ledger

Code	h	Class	r(h)	Recip RA	Recip Dec	Direct RA	Direct Dec
001	(0, 0, 1)	axial	1.000000000000	258.000576203	-21.000692601	268.511403628	-60.519741294
011	(0, 1, 1)	face diagonal	1.048796632255	106.780626622	-75.703939558	277.863629234	-81.531313278
010	(0, 1, 0)	axial	1.339905341976	84.132672287	-29.408562676	80.328748765	-68.983472217
100	(1, 0, 0)	axial	1.836152673337	355.201792089	1.869279271	350.484327659	-6.440864866
101	(1, 0, 1)	face diagonal	1.974587074999	326.873754621	-8.694351580	305.515488300	-53.084389062
111	(1, 1, 1)	body diagonal	2.000023101807	3.082265054	-28.568337611	326.059899967	-71.554434722
01-1	(0, 1, -1)	face diagonal	2.119131490749	81.407975652	-8.126533422	85.646251048	13.370169239
10-1	(1, 0, -1)	face diagonal	2.200890538590	20.583534830	10.955542673	44.766577125	53.904492848
1-1-1	(1, -1, -1)	body diagonal	2.223234046047	348.090586846	28.951893226	22.879798738	74.703177716
1-10	(1, -1, 0)	face diagonal	2.272813025763	322.437786858	18.411205582	318.092690137	52.407301840
110	(1, 1, 0)	face diagonal	2.273307476001	27.349970066	-15.252460507	22.786888326	-56.064996547
1-11	(1, -1, 1)	body diagonal	2.718195812787	304.223019989	7.599043207	293.392122777	-14.204847349
11-1	(1, 1, -1)	body diagonal	2.887171860683	41.881515162	-4.761634499	55.477731998	9.052090062

The reported and recomputed radii agree to a maximum absolute difference of 4.441e-16. The direct and reciprocal celestial directions are alternative metric renderings of the same finite projective address.

18. Complete affine-plane family ledger

Each projective normal defines three parallel affine planes. Every plane contains exactly nine of the 27 Lucas addresses. Compact node labels use m for -1.

Normal code	h	Residue	Count	Nine affine points
001	(0,0,1)	0	9	(-1,-1,0), (-1,0,0), (0,-1,0), (-1,1,0), (0,0,0), (1,-1,0), (0,1,0), (1,0,0), (1,1,0)
001	(0,0,1)	1	9	(-1,-1,1), (-1,0,1), (0,-1,1), (-1,1,1), (0,0,1), (1,-1,1), (0,1,1), (1,0,1), (1,1,1)
001	(0,0,1)	2	9	(-1,-1,-1), (-1,0,-1), (0,-1,-1), (-1,1,-1), (0,0,-1), (1,-1,-1), (0,1,-1), (1,0,-1), (1,1,-1)
011	(0,1,1)	0	9	(-1,-1,1), (-1,0,0), (-1,1,-1), (0,-1,1), (0,0,0), (0,1,-1), (1,-1,1), (1,0,0), (1,1,-1)
011	(0,1,1)	1	9	(-1,-1,-1), (0,-1,-1), (1,-1,-1), (-1,0,1), (-1,1,0), (0,0,1), (0,1,0), (1,0,1), (1,1,0)
011	(0,1,1)	2	9	(-1,-1,0), (-1,0,-1), (0,-1,0), (0,0,-1), (1,-1,0), (1,0,-1), (-1,1,1), (0,1,1), (1,1,1)
010	(0,1,0)	0	9	(-1,0,-1), (-1,0,0), (0,0,-1), (-1,0,1), (0,0,0), (1,0,-1), (0,0,1), (1,0,0), (1,0,1)
010	(0,1,0)	1	9	(-1,1,-1), (-1,1,0), (0,1,-1), (-1,1,1), (0,1,0), (1,1,-1), (0,1,1), (1,1,0), (1,1,1)
010	(0,1,0)	2	9	(-1,-1,-1), (-1,-1,0), (0,-1,-1), (-1,-1,1), (0,-1,0), (1,-1,-1), (0,-1,1), (1,-1,0), (1,-1,1)
100	(1,0,0)	0	9	(0,-1,-1), (0,-1,0), (0,0,-1), (0,-1,1), (0,0,0), (0,1,-1), (0,0,1), (0,1,0), (0,1,1)
100	(1,0,0)	1	9	(1,-1,-1), (1,-1,0), (1,0,-1), (1,-1,1), (1,0,0), (1,1,-1), (1,0,1), (1,1,0), (1,1,1)
100	(1,0,0)	2	9	(-1,-1,-1), (-1,-1,0), (-1,0,-1), (-1,-1,1), (-1,0,0), (-1,1,-1), (-1,0,1), (-1,1,0), (-1,1,1)
101	(1,0,1)	0	9	(-1,-1,1), (0,-1,0), (1,-1,-1), (-1,0,1), (0,0,0), (1,0,-1), (-1,1,1), (0,1,0), (1,1,-1)
101	(1,0,1)	1	9	(-1,-1,-1), (-1,0,-1), (-1,1,-1), (0,-1,1), (1,-1,0), (0,0,1), (1,0,0), (0,1,1), (1,1,0)
101	(1,0,1)	2	9	(-1,-1,0), (0,-1,-1), (-1,0,0), (0,0,-1), (-1,1,0), (0,1,-1), (1,-1,1), (1,0,1), (1,1,1)
111	(1,1,1)	0	9	(-1,-1,-1), (-1,0,1), (-1,1,0), (0,-1,1), (0,0,0), (0,1,-1), (1,-1,0), (1,0,-1), (1,1,1)
111	(1,1,1)	1	9	(-1,-1,0), (-1,0,-1), (0,-1,-1), (-1,1,1), (0,0,1), (0,1,0), (1,-1,1), (1,0,0), (1,1,-1)
111	(1,1,1)	2	9	(-1,-1,1), (-1,0,0), (-1,1,-1), (0,-1,0), (0,0,-1), (1,-1,-1), (0,1,1), (1,0,1), (1,1,0)
01-1	(0,1,-1)	0	9	(-1,-1,-1), (0,-1,-1), (-1,0,0), (1,-1,-1), (0,0,0), (-1,1,1), (1,0,0), (0,1,1), (1,1,1)
01-1	(0,1,-1)	1	9	(-1,0,-1), (-1,1,1), (0,0,-1), (-1,1,0), (0,-1,1), (1,0,-1), (0,1,0), (1,-1,1), (1,1,0)
01-1	(0,1,-1)	2	9	(-1,-1,0), (-1,1,-1), (0,-1,0), (-1,0,1), (0,1,-1), (1,-1,0), (0,0,1), (1,1,-1), (1,0,1)

Normal code	h	Residue	Count	Nine affine points
10-1	(1,0,-1)	0	9	(-1,-1,-1), (-1,0,-1), (-1,1,-1), (0,-1,0), (0,0,0), (0,1,0), (1,-1,1), (1,0,1), (1,1,1)
10-1	(1,0,-1)	1	9	(0,-1,-1), (-1,-1,1), (0,0,-1), (-1,0,1), (0,1,-1), (1,-1,0), (-1,1,1), (1,0,0), (1,1,0)
10-1	(1,0,-1)	2	9	(-1,-1,0), (-1,0,0), (1,-1,-1), (-1,1,0), (0,-1,1), (1,0,-1), (0,0,1), (1,1,-1), (0,1,1)
1-1-1	(1,-1,-1)	0	9	(-1,-1,0), (-1,0,-1), (1,-1,-1), (0,-1,1), (0,0,0), (0,1,-1), (-1,1,1), (1,0,1), (1,1,0)
1-1-1	(1,-1,-1)	1	9	(-1,-1,-1), (0,-1,0), (0,0,-1), (-1,0,1), (-1,1,0), (1,-1,1), (1,0,0), (1,1,-1), (0,1,1)
1-1-1	(1,-1,-1)	2	9	(0,-1,-1), (-1,-1,1), (-1,0,0), (-1,1,-1), (1,-1,0), (1,0,-1), (0,0,1), (0,1,0), (1,1,1)
1-10	(1,-1,0)	0	9	(-1,-1,-1), (-1,-1,0), (-1,-1,1), (0,0,-1), (0,0,0), (0,0,1), (1,1,-1), (1,1,0), (1,1,1)
1-10	(1,-1,0)	1	9	(0,-1,-1), (-1,1,-1), (0,-1,0), (-1,1,0), (0,-1,1), (1,0,-1), (-1,1,1), (1,0,0), (1,0,1)
1-10	(1,-1,0)	2	9	(-1,0,-1), (-1,0,0), (1,-1,-1), (-1,0,1), (0,1,-1), (1,-1,0), (0,1,0), (1,-1,1), (0,1,1)
110	(1,1,0)	0	9	(-1,1,-1), (0,0,-1), (1,-1,-1), (-1,1,0), (0,0,0), (1,-1,0), (-1,1,1), (0,0,1), (1,-1,1)
110	(1,1,0)	1	9	(-1,-1,-1), (-1,-1,0), (-1,-1,1), (0,1,-1), (1,0,-1), (0,1,0), (1,0,0), (0,1,1), (1,0,1)
110	(1,1,0)	2	9	(-1,0,-1), (0,-1,-1), (-1,0,0), (0,-1,0), (-1,0,1), (0,-1,1), (1,1,-1), (1,1,0), (1,1,1)
1-11	(1,-1,1)	0	9	(-1,-1,0), (0,-1,-1), (-1,1,-1), (-1,0,1), (0,0,0), (1,0,-1), (1,-1,1), (0,1,1), (1,1,0)
1-11	(1,-1,1)	1	9	(-1,0,-1), (-1,-1,1), (0,-1,0), (1,-1,-1), (-1,1,0), (0,1,-1), (0,0,1), (1,0,0), (1,1,1)
1-11	(1,-1,1)	2	9	(-1,-1,-1), (-1,0,0), (0,0,-1), (0,-1,1), (1,-1,0), (-1,1,1), (0,1,0), (1,1,-1), (1,0,1)
11-1	(1,1,-1)	0	9	(-1,0,-1), (0,-1,-1), (-1,-1,1), (-1,1,0), (0,0,0), (1,-1,0), (1,1,-1), (0,1,1), (1,0,1)
11-1	(1,1,-1)	1	9	(-1,-1,0), (-1,1,-1), (0,0,-1), (1,-1,-1), (-1,0,1), (0,-1,1), (0,1,0), (1,0,0), (1,1,1)
11-1	(1,1,-1)	2	9	(-1,-1,-1), (-1,0,0), (0,-1,0), (0,1,-1), (1,0,-1), (-1,1,1), (0,0,1), (1,-1,1), (1,1,0)

19. Glossary of native terms

Address space. The integer or finite-field coordinate space in which exact lattice, phase, incidence, and congruence relations are stated.

Physical embedding. The map x = B u that converts an address into a dimensioned Cartesian vector.

AG(3,3). The 27-point three-dimensional affine space over the finite field F3.

PG(2,3). The 13-point projective plane obtained from the 26 nonzero ternary vectors after antipodal equivalence.

Character. A finite Fourier basis function psi_h(u)=omega^(h dot u).

Projective shell. The 13 nonzero conjugate Fourier sectors and their direct or reciprocal physical directions.

Incidence. The relation h dot k=0 mod 3, equivalent to phase invariance of mode h along direction k.

Exterior bivector. An oriented face-area vector such as b cross c.

Reciprocal metric. G^(-1), the quadratic form governing physical phase-gradient magnitudes.

A1 coordinate. The longitudinal sheet coordinate lambda.

A2 coordinates. The tangential difference coordinates xi1 and xi2 with hexagonal norm.

Gluing congruence. lambda-xi1+xi2=0 mod 3, the condition for transformed coordinates to recover an integer address.

Registry. The combined phase and polarity state carried by lambda modulo six.

Circular-section plane. A central plane on which the phase quadratic form is isotropic with value lambda2.

Closure-null identity. An exact relation that removes a nominal degree of freedom and predicts one observable from others.

Valuation lattice. The rank-two integer scale space generated by powers of two and three.

Primitive variable. A state coordinate not algebraically derived from other coordinates in the chosen model.

20. Archival record and reproducibility ledger

Field	Record
Document title	K-Field Fourier Geometry Hidden Architecture
Principal Investigator	P. Dwayne Esterline
Credentials	BS, Huntington University (1999); MBA, Indiana Wesleyan University (2008)
Organization	Krytronx, LLC
Classification	Proprietary Confidential Information and Trade Secret of Krytronx, LLC. All Rights Reserved. Do Not Distribute.
Generation epoch	1784910918
Output PDF	K-Field_fourier_geomery_hidden_architecture_1784910918.pdf
Machine-readable companion	K-Field_fourier_geomery_hidden_architecture_1784910918.json
Source file	K-Field_as_projective_exterior_lattice_hidden_relations_1784897464(1).json
Source schema	kfield.projective_exterior_lattice_hidden_relations.v1
Source epoch	1784897464
Source SHA-256	be17bf271139d0bc33bc8694e6f390cf648ab05a0ec45ae0817c3874a030c809
Report generation	Python ReportLab; built-in core fonts only
Figure generation	Python numerical reconstruction from the exact source ledger
Coordinate units	Cell3 Cartesian coordinates in nanometers; angles in degrees
Finite arithmetic	F3 represented by residues 0,1,2 with -1 equivalent to 2
PDF margins	0.70 in left/right; 0.72 in top; 0.68 in bottom
Table policy	Dynamically normalized column widths; all cells are wrapped ReportLab Paragraphs; repeatRows=1

20.1 Numerical integrity checks

Verification	Maximum residual
B^T B versus source direct Gram	0.000e+00
Inverse Gram versus source reciprocal Gram	0.000e+00
27-node embedding x-Bu/2	0.000e+00 nm
13-state radius reconstruction	4.441e-16
N N^T - (3I+J)	0
Maximum shell closure residual	1.776e-15
Maximum circular-section residual	1.867e-15
First three orbital PCA variance	99.999995172%

Archival conclusion. The finite geometry, physical embedding, reciprocal shell, exterior representation, incidence spectrum, sheet decomposition, circular phase sections, and orbital closure relations are mutually reconstructible from the supplied authoritative data.